Q.1 Solve

A Explain Minimizing Power Usage.

To minimize power usage in an organization, it's essential to implement strategies that optimize energy consumption across all operational aspects. Here's a detailed explanation that can help achieve full marks in a 10-mark question:

1. Implement Virtualization

What it is: Consolidating multiple physical servers into virtual machines hosted on a single server.

Impact: Reduces the number of physical servers required, lowering power and cooling requirements.

Example: VMware or Hyper-V used for server virtualization.

2. Use Energy-Efficient Hardware

Low-Power Components: Deploy energy-efficient CPUs (e.g., ARM processors), SSDs instead of HDDs, and LED displays.

Energy Star Devices: Use certified devices that consume less power while maintaining performance.

Impact: Directly reduces power consumption.

3. Enable Power Management Settings

What it is: Use software settings to enable features like sleep mode, power-saving mode, or auto-shutoff for idle devices.

Impact: Reduces energy wastage when devices are not in use.

Example: Using tools like Windows Power Management or macOS Energy Saver.

4. Optimize Data Centers

Cooling Efficiency:

Adopt free-air cooling or liquid cooling systems.

Increase data center temperatures to ASHRAE recommended levels (~27°C).

Server Utilization: Implement load balancing to prevent underused servers from running unnecessarily.

Impact: Data centers, which are significant power consumers, become more sustainable.

5. Adopt Renewable Energy Sources

Solar Panels or Wind Power: Install renewable energy systems to power IT infrastructure.

Impact: Reduces dependency on fossil fuels and lowers carbon footprint.

Example: Google's data centers powered by renewable energy sources.

6. Conduct Energy Audits

What it is: Regularly assess energy usage to identify inefficiencies in power consumption.

Impact: Highlights areas for improvement and ensures adherence to green strategies.

7. Implement Thin Client Computing

What it is: Use thin clients (low-power terminals) instead of full-fledged PCs for end users.

Impact: Reduces energy usage and extends hardware lifecycle.

Example: Dell Thin Clients in office environments.

8. Replace Traditional Lighting

What it is: Use LED lighting instead of fluorescent or incandescent bulbs.

Impact: Saves up to 75% energy compared to traditional lighting.

9. Educate Employees

Awareness Campaigns: Train employees to turn off devices when not in use, unplug chargers, and minimize unnecessary printing.

Impact: Creates a culture of energy conservation.

10. Use Energy Monitoring Tools

Smart Meters and Dashboards: Track energy consumption in real-time.

Impact: Helps in identifying high-power-consuming equipment and optimizing its usage.

Real-Life Impact

Organizations like Google and Amazon have successfully minimized power usage by implementing advanced cooling systems and running their data centers on renewable energy. Small-scale businesses can achieve similar results by starting with basic measures like enabling power management and switching to efficient hardware.

B What are Smart Buildings and Homes? How does it help in improving environmental Sustainability?

A smart home (or building) is an intelligent home (or building) that maximizes itsenergy efficiency and performance by integrating various building subsystems including communications, entertainment, security, convenience and information systems. Withthe advent of newer technologies and advanced products, intelligent buildings are fast becoming a reality.

Buildings can be considered as energy-intensive systems throughout their whole lifecycle including the design, realization (construction) and support (maintenance and ren ovation) phases (European Commission, 2009).

According to a World Business Council for Sustainable Development estimate(WBSCD, 2008), buildings consume ~ 40% of the world's energy, and almost all building complexes, manufacturing plants, retail stores, hospitals and hotels waste energy.

Buildings represent a complex interdependent system comprising tenants, buildings and building equipment. Energy consumption profiles and concerns differ across locations,

structures, seasons and occupancy levels. For instance, buildings account for about 40% of energy demand, and the worldwide energy consumption for buildings will grow by 45% from 2002 to 2025 (The Climate Group, 2008).

Energy efficiency in buildings could be achieved by adopting best environmentally friendly design and engineering approaches, sustainable procurement processes and conformance to building codes and regulations for energy efficiency. The renovation of existing buildings for greater energy efficiency constitutes sustainable modernization.

IT could be harnessed to improve building energy efficiency and reduce total energy consumption. For instance, energy efficiency in future smart buildings can be gained through smart building management systems and energy control management systems supported by wired and wireless sensors, multimodal interactive interfaces and multimodal context-aware systems (European Commission, 2009).

Many building professionals are unaware of the significance of energy efficiency in buildings and tend to underestimate the contribution of buildings' energy to sustainability. Greater awareness and education are needed to motivate builders and residents to use IT-enabled smart systems to reduce energy consumption and energy bills.

How Smart Buildings and Homes Improve Environmental Sustainability

Energy Efficiency

- Real-time monitoring and automated controls reduce unnecessary energy consumption.
- Smart lighting and HVAC (Heating, Ventilation, and Air Conditioning) systems adjust usage based on occupancy, saving electricity.
- Example: Smart thermostats like Nest can cut energy costs by up to 15%.

Reduction in Carbon Emissions

- Reduced energy consumption leads to lower dependency on fossil fuels, decreasing greenhouse gas emissions.
- Integration of renewable energy sources, such as rooftop solar panels, further cuts carbon footprints.

Water Conservation

- Smart irrigation systems prevent overwatering by analyzing soil moisture levels.
- Automated faucets ensure minimal water wastage in homes and buildings.

Waste Reduction

- Smart homes use sensors to manage waste, such as notifying when bins are full or optimizing recycling processes.

Improved Air Quality

- Smart air purification systems monitor and improve indoor air quality by reducing pollutants, benefiting both health and the environment.

Integration of Sustainable Materials

- Smart buildings often incorporate sustainable construction materials that lower the overall ecological impact.

C Explain Metrics and Measurements in Green Strategies.

Metrics for green IT performance of an organization can be based internal ROI goals and/or on legal reporting requirements. Whilst the ISO 14000 series of standards can provide an excellent starting point for the Key Performance Indicators (KPIs) for green IT, CEMS can be used to automate, measure and report on carbon emissions and the carbon footprint.

Following are some typical KPIs that must be embedded in an organization that is undertaking green strategies.

• **Economic outcome.** Reduce energy consumption by 10% of its current level per year for three years; increase green services (e.g. the addition of one detailed insurance service dedicated to green).

• **Technical.** Use virtualized data servers for all warehoused data; use smart meters to record, repost and control emissions.

• **Process.** Optimize supply chain management to reduce or reengineer individual processes.

• **People.** Train people for green IT at all levels. Telecommute once a week to reduce emissions.

The KPI groups described here can be further expanded in greater detail, and they will have their own nuances depending on the dimension to which they belong. For example, a KPI that is entirely focussed on carbon reduction irrespective of cost considerations may not be acceptable to an organization's economic dimension. Alternatively, a technologically advanced energy-efficient cooler might use less energy and hence have lower operational cost, but the capital expenses towards such a cooler will be part of the economic dimension. Thus, KPIs should regularly tie business efficiency with carbon reduction efficiency. Savings in carbon related to an organization's various aspects, such as production, sales and marketing, research and development and administration, need to be related to savings in costs.

D Write a short note on Green Computing and the Future.

Green Computing refers to the practice of designing, using, and disposing of computing resources in an environmentally sustainable manner. It focuses on minimizing energy consumption, reducing electronic waste, and promoting eco-friendly technologies. Key approaches include using energy-efficient hardware, optimizing software, virtualization, cloud computing, and adopting renewable energy sources in IT infrastructure.

Significance:

Green computing helps reduce the carbon footprint of IT systems, conserve energy, and promote recycling, aligning with global efforts to combat climate change.

The Future of Green Computing

- AI and Automation: AI-powered systems will optimize energy usage in real-time, enhancing resource efficiency.
- Renewable Energy Integration: Data centers and IT systems will increasingly adopt solar, wind, and other renewable energy sources.
- Edge Computing: Reducing data transmission to centralized servers will cut power usage and improve efficiency.
- Circular Economy: Enhanced recycling of e-waste and the use of biodegradable components will reduce environmental impact.
- Government Policies: Stricter regulations and incentives will drive organizations to adopt greener IT practices.

Q.2

A Explain Life Cycle of Device or Hardware.

A green device cannot be built by just having an additional step in the life cycle of the device. There needs to be a concerted effort at every stage of the device life cycle – from the moment the device is conceived, to its development, to the time when it is used and recycled or disposed (i.e. from cradle to grave). Each stage of the cycle has varying levels of impact on the environment. In this section, we will look into each stage that a computer device goes through and discuss green considerations for each stage. The typical life cycle of a device, shown in Figure 2.1, consists of five stages:

1. Design.
2. Manufacture and facilities.
3. Packaging and transportation.
4. Usage.
5. Reuse or disposal.

Figure 2.1 Life cycle of a device.

In the design stage, the idea is conceptualized and the device is designed, prototyped and tested. Though this stage does not have direct impact on the environment, design stage decisions such as those regarding architecture, constituent components or materials and layout have a huge effect on the environmental impact of other stages. Hence, it is important to design devices so as to keep their environmental impact as low as possible during subsequent stages whilst meeting performance and other requirements. We will look at a possible process improvement that will help design a green device.

Manufacturing

The manufacturing process is one of the main sources of environmental impact in the life cycle of a device. Manufacturing processes are resource intensive and consume a lot of raw materials, water and energy; they create many different categories of waste, some which are toxic. For instance, according to a UN report, production of a desktop per

sonal computer (PC) consumes 240 kg of fossil fuels, 22 kg of chemicals (which includes hazardous chemicals) and 1.5 tonnes of water (United Nations Environment Programme, 2004). It is very important to minimize the use of environmentally sensitive substances and reduce the amount of harmful waste.
The EPEAT (Electronic Product Environmental Assessment Tool, n.d.) ratings system serves as a good guideline for understanding how green a device is. EPEAT grades electronic devices and computer systems into three grades (bronze, silver and gold), helps customers identify how green a device is and helps them to make informed choices.
The rating system takes into account the following aspects:

- Reduction or elimination of environmentally sensitive materials;
- Material selection;
- Design for end of life;
- Device longevity or life extension;
- Energy conservation;
- End-of-life management;
- Corporate performance;
- Packaging.

Table 2.1 presents a brief summary of the hazardous chemicals used in manufacturing various electronic devices and their effects on humans.
The impact of some of those key materials is as follows:

- **Lead** is generally used in printed circuit boards and in glass monitors. The pathological effects of lead are most prominent in the nervous system, hematopoietic system and kidneys. Excessive exposure to lead (Pb) results in clinical toxicity. There has been extensive research to find a replacement for lead (Norwegian University of Science and Technology, 2010). There are also on-going efforts to recover lead from the e-waste (Duffy, 2008).
- **Cadmium** is generally used in low-temperature soldering, plating for corrosion protection, colorants in plastics and contact buttons in relays (Penica and Hilty, 2004). Cadmium when ingested affects the liver and kidneys. Research is on-going to produce alternatives for its various uses.
- **Mercury** is mainly used in monitors and batteries. Mercury is a well-known pollutant (US Geological Survey, 2000), and it affects the immune system, alters genetic and enzyme systems and damages the nervous system. New light-emitting diode (LED) backlit technologies have been developed which make monitors mercury free. Mercury use in batteries has also been reduced considerably over the past few years.
- **PBDEs and PBBs** are used as flame retardants (Wisconsin Department of Natural Resources, 2011). These materials are ubiquitous environmental pollutants; they are extremely toxic and are banned in

many countries. Efforts are on-going to remove these materials from the manufacture of computers.

• **Arsenic** is used in glasses and in manufacturing semiconductors. Arsenic interferes with cellular longevity by allosteric inhibition of an essential metabolic enzyme. Many arsenic prevention and recovery methodologies have been devised, and these need to be actively deployed to reduce arsenic usage in the manufacturing process.

• **PVC** is one of the most commonly used plastics worldwide. It has been used extensively in the manufacture of computer parts. Vinyl chloride, one of the main ingredients of PVC, is a carcinogen and also affects the human reproductive system (US Environmental Protection Agency (EPA), 2011). Of late, many companies have eliminated the use of PVC in their devices.

Packaging

Packaging and transportation also contribute to the carbon footprint in a device's life cycle. The two main contributors in this segment are the materials used for packaging and the carbon footprint of the vehicles used in transportation.

The following are general recommendations for designing suitable packaging (US Environmental Protection Agency, 2011):

1. Packaging materials should be recyclable.
2. The amount of packaging materials should be kept at a minimum.
3. Various materials used in packaging should be easily separable to ease the recycling process.
4. Adhesive use should be reduced by using folds and tabs instead.
5. All the additives, coatings and inks that get added to the package should be eco-friendly.
6. Printed documentation can be avoided wherever possible. Instead, documentation could be provided through the device's Web page.

In addition, freight transportation leads to a lot of carbon emission. Along with making the transport system efficient, the design of the device should facilitate smarter transportation. For example, the smaller the device, the more of them can be transported in a given space, thus reducing the number of vehicles required to transport them from manufacturing facilities to distribution centres.

Usage

The amount of energy consumed by different kinds of electronic devices varies widely, and ways to reduce their energy use vary widely too, hence we will look into different device categories separately, and look at ways to reduce energy usage in notebook computers, desktop computers, servers, mobile devices and other special devices.

1. Notebook Computers
Energy Usage: Notebook computers are designed to be energy-efficient compared to desktops, consuming around 20-50 watts during operation.

Ways to Reduce Energy Use:
Enable Power-Saving Modes: Use features like sleep mode and energy saver settings.
Adjust Screen Brightness: Lower the brightness to conserve battery.
Efficient Charging Practices: Unplug chargers when batteries are fully charged.
Close Unused Applications: Minimize processing power demands.

2. Desktop Computers
Energy Usage: Desktops consume higher energy, typically 60-300 watts, due to larger components and constant AC power supply.
Ways to Reduce Energy Use:
Upgrade to Energy-Efficient Hardware: Use SSDs and energy-efficient CPUs.
Replace CRT Monitors: Use LED or LCD monitors to reduce power consumption.
Virtualization: Consolidate multiple workloads onto fewer systems.
Enable Sleep and Hibernate Modes: Automatically reduce power during idle periods.

3. Servers
Energy Usage: Servers are high-energy consumers, with large-scale data centers consuming megawatts of power annually.
Ways to Reduce Energy Use:
Virtualization and Cloud Computing: Optimize resource allocation and consolidate workloads.
Efficient Cooling Systems: Use advanced cooling techniques, like liquid or free-air cooling.
Energy Monitoring Tools: Continuously monitor and optimize server performance.
Renewable Energy: Power data centers with solar or wind energy.

4. Mobile Devices

Energy Usage: Mobile devices like smartphones and tablets consume 2-6 watts when charging.

Ways to Reduce Energy Use:
Fast-Charging Awareness: Avoid overcharging and unplug devices once fully charged.
Disable Unused Features: Turn off Bluetooth, GPS, and Wi-Fi when not needed.
Use Battery-Saver Mode: Limit background app activities.

5. Special Devices (Printers, Routers, etc.)
Energy Usage:
Printers consume 100-300 watts during operation and 1-5 watts in standby mode.
Routers use about 5-15 watts continuously.
Ways to Reduce Energy Use:
Printers: Use duplex printing to save paper and energy; turn off when not in use.
Routers: Use energy-efficient models and schedule downtime (e.g., turning off at night)

B What is mean by Green IT? Explain holistic approach to Greening IT.

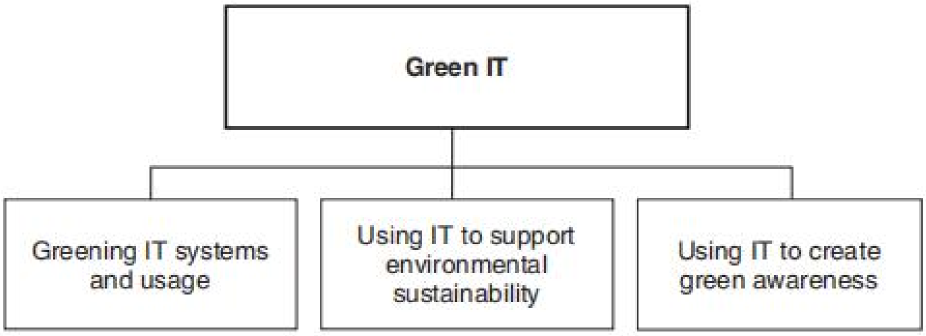

Figure 1.1 Green IT dimensions.

Green IT is an umbrella term referring to environmentally sound information technologies and systems, applications and practices. It encompasses three complementary IT-enabled approaches to improving environmental sustainability (Murugesan, 2008) (see Figure 1.1):

1. the efficient and effective design, manufacture, use and disposal of computer hardware, software and communication systems with no or minimal impact on the environment;
2. the use of IT and information systems to empower – that is, support, assist and leverage – other enterprise-wide environmental initiatives and
3. the harnessing of IT to help create awareness among stakeholders and promote the green agenda and green initiatives.

To comprehensively and effectively address the environmental impacts of IT, we must adopt a holistic approach that addresses the problems along these six complementary directions (Murugesan, 2008) (see Figure 1.3):

1. **Green design.** Design energy-efficient and environmentally sound components, computers, servers and cooling equipment.
2. **Green manufacturing.** Manufacture electronic components, computers and other associated subsystems with minimal or no impact on the environment.
3. **Green use.** Reduce the energy consumption of computers and other information systems, and use them in an environmentally sound manner.
4. **Green disposal.** Refurbish and reuse old computers, and properly recycle unwanted computers and other electronic equipment.
5. **Green standards and metrics.** These are required for promoting, comparing and benchmarking sustainability initiatives, products, services and practices.

6. **Green IT strategies and policies.** These effective and actionable strategies and policies add value and focus on both short- and long-term benefits. These are aligned with business strategies and practices, and are key components of greening IT.

the entire life cycle of a computer, server and storage system could be made greener, reducing their GHG emissions and carbon footprint and minimizing or eliminating toxic materials used and/or released to the environment. Chapter 2 discusses environmental issues arising from electronic devices, offers a range of solutions to address this problem and highlights best practices in each of the life cycle phases

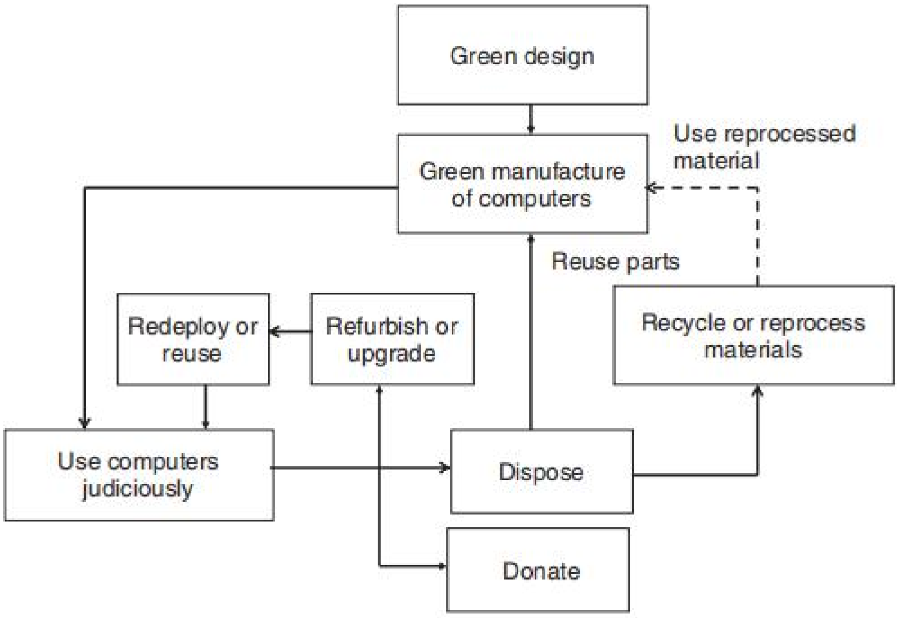

Figure 1.4 Greening computer's entire life cycle.

1.5.2 The Three Rs of Green IT

Unwanted computers, monitors and other hardware should not be thrown away as rubbish, as they will then end up in landfills and cause serious environmental problems. Instead, we should refurbish and reuse them, or dispose them in environmentally sound ways.

Reuse, *refurbish* and *recycle* are the three 'Rs' of greening unwanted hardware.

• **Reuse.** Many organizations and individuals buy new computers for each project or once every 2–3 years. Instead, we should make use of an older computer if it meets our requirements. Otherwise, we should give it to someone who could use it in another project or unit. By using hardware for a longer period of time, we can reduce the total environmental footprint caused by computer manufacturing and disposal.

- **Refurbish.** We can refurbish and upgrade old computers and servers to meet our new requirements. We can make an old computer and other IT hardware almost new again by reconditioning and replacing some parts. Rather than buying a new computer to our specifications, we can also buy refurbished IT hardware in the market. More enterprises are now open to purchasing refurbished IT hardware, and the market for refurbished equipment is growing. If these options are unsuitable, we can donate the equipment to charities, schools or someone in need, or we can trade in our computers.
- **Recycle.** When we cannot refurbish or otherwise reuse computers, we must dispose of them in environmentally friendly ways by depositing them with recognized electronic recyclers or electronic waste (e-waste) collectors. E-waste – discarded computers and electronic goods – is one of the fastest-growing waste types and poses serious environmental problems. The United Nations Environment Program estimates that 20–50 million tons of e-waste are generated worldwide each year, and this is increasing. IT hardware contains toxic materials like lead, chromium, cadmium and mercury. If we bury IT hardware in landfills, toxic materials can leach harmful chemicals into water ways and the environment. If burned, they release toxic gases into the air we breathe. So if e-waste is not discarded properly, it can harm the environment and us. Waste electrical and electronic equipment (WEEE) regulations aim to reduce the amount of e-waste going to landfills and increase recovery and recycling rates.

Q.3

A Explain any five software sustainability attribute and their metrics .

To evaluate the sustainability performance of a given software system, we must first outline what attributes of the system we want to improve, and examine their impact and benefits
Though many may not recognize it, these software attributes bring environmental, social as well as economic benefits.

- **Modifiability and reusability:** Less effort in introducing changes minimizes the resources used by the development team as well as the amount of waste, and hence reduces the carbon footprint.
- **Portability:** Software that can run on different environments increases flexibility and minimizes e-waste because it can run on existing hardware, for example by installing a less demanding OS like Linux to run the application. Portability also reduces the costs of adoption, and therefore the application is accessible to more users.
- **Supportability:** Less effort in configuring and operating the system minimizes the use of physical resources such as books and training required to support the system.
- **Performance:** Increased performance with even less resources minimizes e-waste by allowing the system to be deployed on older computers.
- **Dependability:** A system that functions reliably and as expected minimizes the need for additional resources because users do not have to deal with recurring unplanned tasks or emergency maintenance.
- **Usability:** By reducing the usability barrier, the system is available to a broader cross section of users. Also, better usability of a system reduces the amount of training required, thus minimizing the use of books, travelling and other resources.
- **Accessibility:** It allows the system to be used by minorities, the elderly, people with disabilities, non-English-speaking communities and the illiterate population.
- **Predictability:** A team that develops software in a predictable manner has to deal with fewer emergencies and unplanned activities, therefore minimizing its usage resources (office resources, computer equipment, electricity, etc.).
- **Efficiency:** A more efficient software team optimizes the use of physical resources.
- **Project carbon footprint:** Decreasing the software team’s carbon footprint directly reduces the amount of greenhouse gas emitted.

Of all the attributes, modifiability and reusability are perhaps the most common in software projects, because they directly affect the programmer's productivity. For instance, in object-oriented systems it is a common practice (Martin, 1994) to measure and minimize the degree of dependency among classes, as highly interdependent classes are difficult and risky to change.

4.5.1 Modifiability and Reusability

To find the distance from main sequence, a prime measure of modifiability, we first need calculate *instability* and *abstractness* – both are measures of reusability

Instability = Efferent Couplings/(Afferent Couplings + EfferentCouplings)

where Afferent Couplings is the number of classes outside a package that depend upon classes within the package, and Efferent Couplings is the number of classes inside a package that depend upon classes outside the package.

Instability ranges from 0 to 1, where 0 means the package is maximally stable and 1 means the package is maximally unstable.

Abstractness is the number of abstract classes in a given package divided by the number of concrete classes in a given package. It ranges from 0 to 1, where 0 means the package is completely concrete and 1 means it is completely abstract.

Distance from Main Sequence = Abstractness + Instability − 1

It ranges from 0 to 1, where 0 means the package has the ideal balance and 1 means the package requires redesign and refactoring. These metrics help us to identify whether the package needs to be optimized in the next release.

4.5.3 Supportability

Another objective is to minimize the amount of post-deployment support required by the system, which would in turn reduce the amount of resources needed. One could use metrics already gathered by the system support team, such as number of calls and average time to resolve tickets. For new systems one can use a metric devised from the usability study, called the support rate. For further information on the usability study, refer to Section 4.6.1.

Support Rate = User’s Questions That Required Assistance/Number of Minutes System Was Used in a Usability Study Session.

4.5.4 Performance

A measure of performance is relative to the objective of the system. Since our goal is to improve user productivity, we use the relative response time metric captured during the usability study.

Relative Response Time = Number of Tasks with an Unacceptable Response Time/Total Number of Tasks Tested.

4.5.5 Dependability

The goal is essentially to minimize defects, which are costly from both business and environmental standpoints, because it causes more resources to be wasted. We use three common quality assurance metrics to evaluate dependability: Defects Density (the amount of defects compared to the system’s size), Testing Efficiency and Testing Effectiveness.

As the use of Defect Density alone could be misleading if the testing process is faulty,

we also use the Testing Efficiency and Testing Effectiveness metrics.

- Defect Density = Known Defects (Found but Not Removed)/Lines of Code (LOC);
- Testing Efficiency = Defects Found/Days of Testing;
- Testing Effectiveness = Defects Found and Removed/Defects Found.

4.5.9 Efficiency

Project efficiency is measured by the effort towards deliverables that add direct value to the customer (e.g. coding and manuals) as opposed to project-related effort (e.g. infrastructure maintenance and project management). The efficiency can be represented in hours or points, depending whether your project is based on a waterfall or agile methodology

4.5.10 Project’s Carbon Footprint

To measure the project’s carbon footprint, rather than quantifying the amount of all the resources used during a project, which may be hard to do, one can track the activities that can potentially cause a negative impact on the environment, such as the following:

- Work-from-home days;
- Long-haul round trips;
- Number of conference calls;
- Number of offsite meetings.

B Explain Green Network and communication.

1.6.6 Green Networking and Communications

Networks and communications play more significant roles than ever before, and facilitate data transfer and sharing. They enable us to communicate and share information, shop, learn and socialize online, make our work environment smarter and do many other things. The demands on communication networks, wired and wireless, have been constantly increasing, and as a result the energy consumption of communication systems has increased considerably. Traditional networking systems and communication protocols are not particularly designed for energy efficiency. Thus they have some negative impact on the environment.

Green networking refers to ways of minimizing networks' impact on the environment using energy-efficient networking technologies, protocols and products and minimizing resource use whenever possible. Green networking practices include the following:

- Using newer, more energy-efficient techniques, technologies and products;
- Upgrading older equipment with newer, greener networking gears;
- Employing smart systems, user management and energy conservation across IT networks to increase energy efficiency;
- Substituting telecommuting, remote administration and video conferencing for travel.

7.1.2 The Challenge of Next-Generation Networks

NGNs pose a challenge in the provision of energy-efficient solutions given their transportation of data with a range of QoS requirements and tolerance of lower than optimum services. Applications which may be transmitted across NGNs include:

1. those with real-time interactivity requirements and ability to accommodate slight loss (e.g. voice);
2. those with real-time interactive requirements and inability to cope with loss (e.g. online multiplayer games);
3. those without real-time requirements but which cannot cope with any loss (e.g. file transfer) and
4. those without real-time requirements and with ability to cope with slight loss (e.g. video download).

7.1.3 Benefits of Energy-Efficient Networks

Energy efficiency strategies are subsequently being developed for use in NGNs, and energy constraints and efficiency objectives of telecommunications operations influence the management strategies deployed. When power saving is applied in a notebook computer, for example, the display screen backlight dims as part of a battery conservation technique. In contrast, when power saving is applied in a wireless sensor network, an intermediary node may have functionality to 'shut down' so that only limited probe packets are distributed to determine its need to 'awaken' and become partially or fully functional.

7.1.4 Objectives of Green Networking

Across the range of domains, objectives of green network communication and management solutions include the following:

- Minimizing the carbon footprint of delivery networks;
- Improving operational sustainability in wireless networks;
- Minimizing the financial cost for operators to transmit;
- Allowing application QoS to be achieved within network resource constraints;
- Reducing load on the network and hence per transaction power consumption;
- Removing the digital divide between urban and rural areas;
- Contributing to industrial standards.

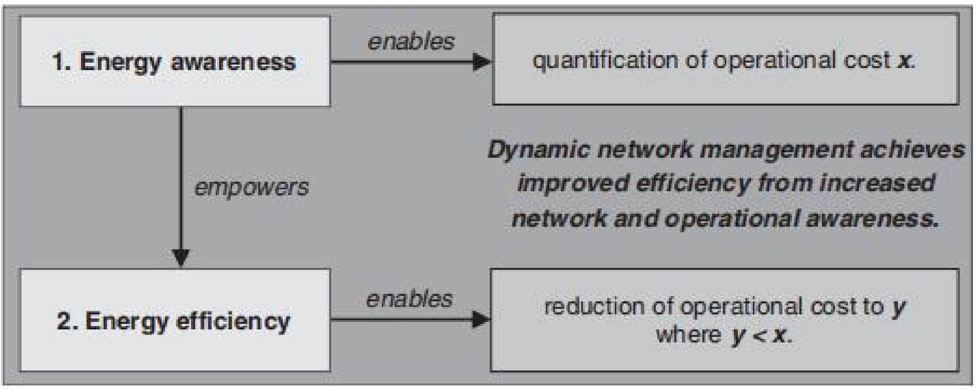

Figure 7.2 Components of next-generation green networks.

7.1.5 Core Components in Green-Networking Technology

To achieve sustainable and lower cost services through green technology, networks require two core capabilities (Figure 7.2): energy awareness and energy efficiency. *Energy awareness* refers to the network's ability to quantify energy cost per packet, and identify if power constraints are becoming a limiting force on its ability to operate or if carbon emissions are increasing above a threshold. *Energy efficiency* describes the network's ability to reduce carbon contributions from those incurred prior to the application of energy efficiency and extend the network's lifetime whilst maintaining QoS. Energy awareness empowers networks with capabilities, absent previously, for effi-ciency objectives. Empowering this ability enables reorganization of the way in which communication occurs, including the operations at each stack layer and management function in general. Network communication protocol standards traditionally focus on achievement of reliability for applications with stricter QoS requirements or a faster response time for those without. Mechanisms for improved efficiency should, however, be incorporated in a cross-layer approach in next-generation green networks (NGGN) such that these objectives may continue to be achieved in parallel with optimized operation.

Q.4

A Explain Data Centre Facility Infrastructure: Implications for Energy Efficiency.

Data centres range from closet-sized operations to megasized facilities, and their function at its simplest is to store, process and move information. For efficient and effective service provisioning, data centre IT infrastructure needs to be carefully designed. However, from a facilities perspective, IT infrastructure is simply a power-consuming, heat-producing 'critical black box' whereby the energy used for providing services is dissipated as heat by IT infrastructure – processors, memory, input–output systems and other components.
Consequently, the two main functions of the supporting infrastructure are as follows:
1. Ensure that the IT and facilities are supplied with power at all times.
2. Keep the data centre at the required temperature by removing the heat generated.
Throughout this section, we discuss the energy implications of these objectives. In focussing on the supporting infrastructure, we assume that all electrical power delivered to the IT equipment is related to 'useful work'; of course, as alluded to in Section 5.2, this is an oversimplification with underutilized systems consuming significant power in legacy data centres.

Here is an overview of key infrastructure components and their implications for energy efficiency:

1. Power Infrastructure
Key Components: Power distribution units (PDUs), uninterruptible power supplies (UPS), and backup generators. Implications for Energy Efficiency:
Efficient Power Supply: Use energy-efficient UPS systems with high conversion efficiency (>95%) to reduce energy losses.
Power Distribution: Deploy modular PDUs that reduce power loss during distribution.
Renewable Energy Sources: Incorporate solar, wind, or hydropower for sustainable energy supply.

2. Cooling Systems
Key Components: Air conditioners, chillers, fans, and liquid cooling systems.

Implications for Energy Efficiency:
Free-Air Cooling: Use outside air for cooling when the external temperature permits, reducing reliance on mechanical cooling.
Hot and Cold Aisle Containment: Physically separate hot air from cold air to optimize cooling efficiency.
Liquid Cooling: Implement liquid cooling systems that transfer heat more efficiently than traditional air cooling.
AI Optimization: Use AI-driven systems to monitor and dynamically adjust cooling requirements based on server workloads
.
3. IT Equipment
Key Components: Servers, storage devices, and networking equipment.
Implications for Energy Efficiency:
Virtualization: Consolidate workloads onto fewer physical servers, reducing energy consumption.
Energy-Efficient Servers: Deploy servers with energy-efficient processors and components.
Storage Optimization: Use tiered storage systems that move less-used data to lower-power storage devices.

4. Network Infrastructure
Key Components: Switches, routers, and cables.
Implications for Energy Efficiency:
Energy-Efficient Networking Devices: Use switches and routers designed to consume less power.
Topology Optimization: Minimize unnecessary networking equipment through better design of data flow paths.

5. Building Infrastructure
Key Components: Lighting, insulation, and structural design.
Implications for Energy Efficiency:
LED Lighting: Use LED lights with motion sensors to minimize unnecessary energy use.
Thermal Insulation: Improve building insulation to maintain stable internal temperatures and reduce cooling demand.
Green Building Materials: Construct with materials that have a lower environmental impact.

6. Energy Monitoring and Management Systems
Key Components: Smart meters, energy dashboards, and predictive analytics tools.
Implications for Energy Efficiency:

Real-Time Monitoring: Track energy usage across systems to identify inefficiencies.
Predictive Analytics: Use AI to forecast energy demands and optimize resource allocation.
Power Usage Effectiveness (PUE): Aim for a PUE value close to 1, indicating efficient energy use (e.g., Google's data centers achieve ~1.1 PUE).

7. Renewable Energy Integration
Key Components: Solar panels, wind turbines, and energy storage systems.
Implications for Energy Efficiency:
Carbon Neutrality: Using renewable energy reduces dependency on fossil fuels and aligns with sustainability goals.
On-Site Generation: Solar or wind installations can directly power IT equipment, reducing transmission losses.

Challenges to Achieving Energy Efficiency
High initial costs for efficient equipment and renewable energy integration.
Limited expertise in designing energy-optimized data centers.
Balancing performance demands with energy-saving goals.

B What are the steps in developing Green IT strategies? Explain in detail.

8.6 Steps in Developing a Green IT Strategy

After giving due consideration to the wide-ranging organizational factors influencing the development of green IT strategy, this section describes the actual steps in developing a green IT strategy. The resulting strategy document may differ depending on the type, size, location and industrial vertical, and those variations will have to be factored in.

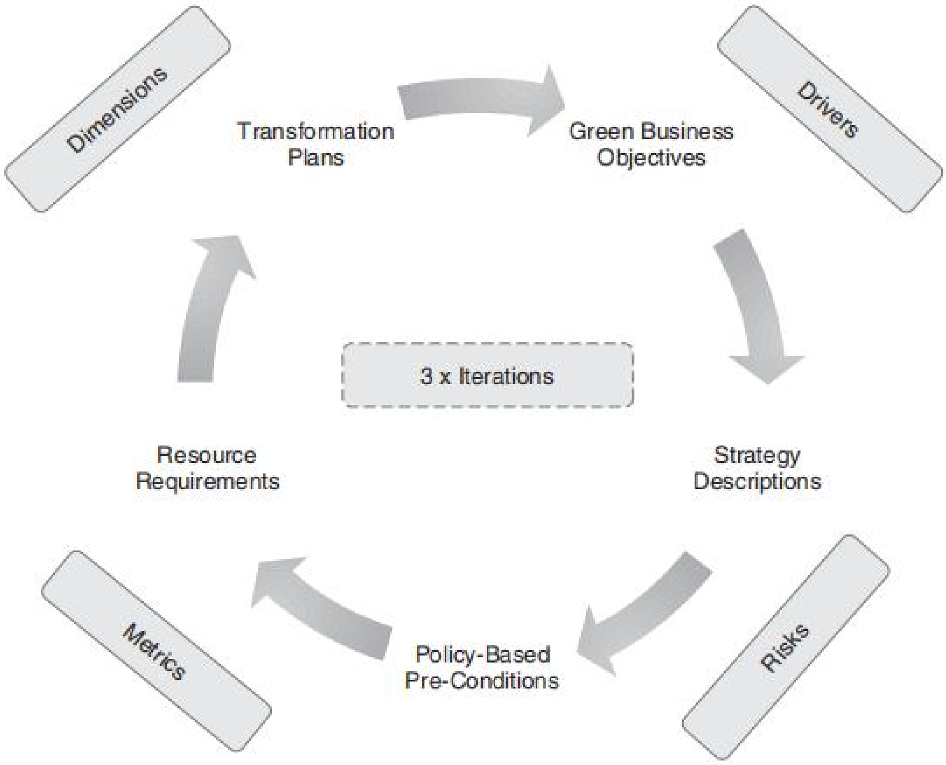

Figure 8.5 Steps in developing a green IT strategy (www.methodscience.com).

The following are the major steps that can be executed iteratively (three times) to produce a green IT strategy (see Figure 8.5).

• **Aligning green IT strategy with business objectives.** Green business objectives are the core objectives for a business undertaking green transformation. As discussed in this chapter, these objectives need to be aligned with an organization's carbon and business goals. The key drivers impacting the organization will provide an understanding of the business objective in becoming green. For example, if cost reduction is the key driver, then that will be reflected in the business objective listed in the green IT strategy – and corresponding metrics applied to it.

• **Strategy descriptions.** This section of the strategy will detail the approach the organization will take in becoming green. The green transformation dimension (economic, technical, process and people) will provide input into the strategy description. The length of time required for implementation (for example, three years) will also come into play here. Organizations can easily take 1–3 months (or more, depending on an organization's size) to develop their green IT strategy descriptions. Strategy descriptions lead to green enterprise transformation projects that influence the organization. This also requires an understanding of the current strategic plan if one exists. Strategy descriptions can be based on SMART (specific, measurable, attainable, realistic and timely) goals (The Climate Group 2008).

• **Transformation plan and timeline.** Developing a green enterprise transformation plan is the final and important step in developing a green IT strategy. A *transformation plan* is a project plan that contains tasks, roles and deliverables together with a timeline for delivery. Usually, this transformation project plan provides the roadmap for transformation. This plan is generally divided into two parts – a high-level roadmap that identifies major areas of work, deliverables and timelines, which can be followed by a detailed, task-by-task project plan that makes use of all known project and programme management techniques.

• **Iterations and risks.** Development of a green IT strategy should not be a unidirectional process. Instead, it should be developed as an iterative process going through all drivers, dimensions, risks and metrics more than once, as depicted in Figure 8.5. All influencing factors, such as the drivers, dimensions, metrics and risks, are also summarized in Figure 8.4. Typically, you may need three iterations to arrive at a final, comprehensive and actionable green IT plan. These iterations, sometimes ranging over a period of 3–6 months, may also include observing industry trends and new developments with respect to green IT. The green policies have to be revised based on these trends. These iterations also indicate an approach to implementing the policies – which should also be based on iterations. Iterative refinements of the policies are expected during practice. However, the concepts of continuous improvements in processes, people and technologies that provide impetus to a lean business initiative also apply here.

Q.5

A Explain readiness and capability in Green IT.

G-readiness refers to the maturity of environmental considerations, whether they're part of a coherent set of IT management policies, they have been adopted into IT human and managerial operational practices or they have been built into concrete IS. It demonstrates the comparative levels of green IT development among enterprises and serves as a benchmark for understanding IT's contribution to support an enterprise's environmental sustainability strategy. G-readiness consists of five components: *attitude*, *policy*, *practice*, *technology* and *governance*. The attitude dimension of G-readiness is an element of the IT human infrastructure and represents the green IT input capability. The policy and governance components are elements of IT managerial capability and represent its transformational capability. The technology and practice components are elements of the IT technical infrastructure and represent its output capability.

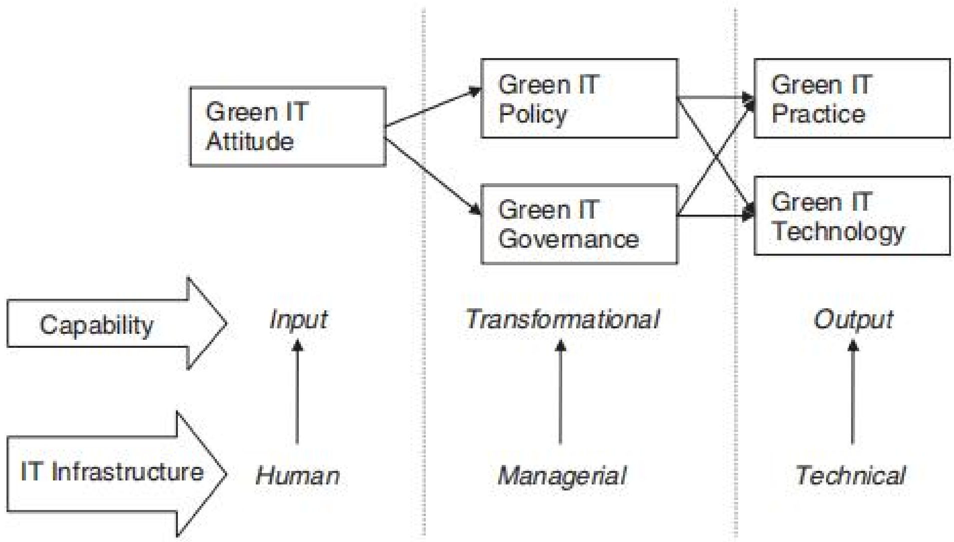

Figure 10.2 The nomological structure of G-readiness.

10.3.1 Green IT Attitude

An *attitude* represents an enduring positive or negative feeling about some object or issue (Eagly and Chaiken, 1993). Attitudes are learned dispositions and are often the result of experiences. In the context of climate change, two major attitudes are prevalent. Most submit that the climate is changing at an alarming rate and that human (including business) activities are the main causes of the change (IPCC, 2007; Stern, 2008). On the other hand, some are sceptical about the causes and impacts of climate change (Royal Society, 2007).

This group maintains that the climate is always changing and its impact is not as severe as most believe. Correspondingly, organizations are likely to adopt very different attitudes at the corporate level in dealing with eco-sustainability, and these differing attitudes will impact their expectations of green IT (CFO, 2009; Hart, 1997; Info-Tech, 2008). *Green IT attitude* is defined as an organization's IT human infrastructure's sentiment towards climate change, and refers to the extent to which IT people are aware and concerned about the impact (both positive and negative) of IT on eco-sustainability. The attitude of IT people and managers towards environmental sustainability and the role of IT is a key factor in initiating actions to green IT (Gartner, 2008). Having a favourable attitude towards green IT is very important – it precedes the development of green IT transformational capability. Thus, the following proposition emerges:

Proposition 1: Firms with IT people and managers who believe that environmental issues are important are more likely to have a relatively well-developed green IT policy, and a relatively matured green IT governance mechanism.

10.3.2 Green IT Policy

Organizations should develop their green IT policy aligned with its overall environmental policy and initiatives. *Green IT policy* encompasses the frameworks the organization puts in place to apply environmental criteria in IT-related activities. It defines the extent to which green issues are encapsulated in an organization's procedures guiding the sourcing, use and disposal of the IT technical infrastructure, the activities of the IT human infrastructure and the use of IT in the wider enterprise (Gartner, 2008; Olson, 2008).

The maturity of green IT policy reflects whether environmental considerations are systematically permeating the IT activity value chain and are repeatable, or are disorganized and based on uncoordinated efforts. Policy captures an organization's intent to green IT.

However, not all policies are expected to be smoothly implemented, nor are all practices expected to be policy led. Thus we arrive at the following propositions:

Proposition 2: Firms with a relatively well-developed green IT policy are more likely to have greener technical IT infrastructure.

Proposition 3: Green IT practices are more likely to flourish in firms that have a relatively well-developed green IT policy.

10.3.3 Green IT Governance

Green IT governance is the operating model that defines the administration of green IT initiatives and is closely related to the policy construct. Roles, responsibilities, account ability and control for green IT initiatives need to be clearly established. Businesses should decide whether the responsibility for green IT initiatives should be delegated to the CIO or the environmental sustainability manager (CFO, 2009; Gartner, 2008). For example, in ANZ Bank and Deloitte, IT leads green IT initiatives, whilst in some other institutions, IT's role is restricted to providing either tools or insights (Gartner, 2008). Deloitte's green IT operating model is based on a three-step process of planning sustainability measures, implementing and tracking the measures and addressing green organizational change (Deloitte, 2010). Green IT governance also includes allocating budget and other resources to green IT initiatives and defining metrics for assessing the impacts of green IT initiatives. Indeed, governance capabilities will require standard administrative processes for developing green IT initiatives to be put in place. This leads to the following two propositions:

Proposition 4: Green IT practices are more likely to flourish in firms with a relatively well-developed green IT governance mechanism.

Proposition 5: Firms with relatively well-developed green IT governance are more likely to have a high level of green IT technical infrastructure.

10.3.4 Green IT Practice

Green IT practice pertains to the actual application and realization of eco-sustainability considerations in IT infrastructure sourcing, operation and disposal. Organizations are likely to vary in the practice of analysing the green track record of IT hardware, software and services providers (CFO, 2009). They are also likely to vary in their practice in operating the IT and network-critical physical infrastructure in data centres and beyond data centres throughout the organization in an eco-friendly manner (Accenture, 2008; CFO, 2009; Velte, Velte, and Elsenpeter, 2008). For example, some are enforcing the Advanced Configuration and Power Interface (ACPI) to slow down processors. In 2005, IBM USA's telework programme involved over 20 000 employees, saving more than 5 million gallons of fuel and avoiding more than 50 000 tons of CO2 emissions. A number of companies either recycle their IT hardware at the end of its life or dispose of it in an environmentally friendly way (CFO, 2009; Mitchell, 2008). For instance, Deloitte's green IT practices involve 'replacing traditional computers with thin laptops (Samson, 2008), for new data centres, and introducing application centralization and platform

standardization’. Green IT practices would contribute positively to greening the IT technical infrastructure.
Proposition 6: Firms that apply eco-considerations in IT infrastructure decisions are more likely to have a green IT technical infrastructure.

10.3.5 Green IT Technology
The *technological* dimension refers to technologies and IS for
(i) reducing the energy consumption of powering and cooling corporate IT assets (such as data centres),
(ii) optimizing the energy efficiency of the IT technical infrastructure,
(iii) reducing IT-induced greenhouse gas emissions,
(iv) supplanting carbon-emitting business practices and
(v) analysing a business’s total environmental footprint

For example,
SAP’s Recycling Administration Application can help organizations to meet regulatory reporting and documentation requirements, manage recycling declaration and payment processes more efficiently and reduce the risk and cost of environmental reporting. The SAP Environmental Compliance application is designed to help ‘organizations ensure compliance with environmental laws and policies and reduce associated costs, efforts and risks on (the) plant and corporate level. It streamlines all environmental processes by seamless integration with operations control data, production control systems, and components from SAP software for environment, health and safety, enterprise asset management, materials management, the SAP Manufacturing Integration and Intelligence (SAP MII) application, business intelligence and knowledge management’.

B How the Communication and Social Media play an important role in managing Green IT.

Communication to stakeholders and stakeholder participation are keys to the success of green initiatives. Among other channels, social media can be used for communicating among stakeholders green initiatives, their progress and their benefits as well as to get their feedback and to engage and involve them. Judiciously using these technologies allows you to present your authoritative stance on the advantages of implementing green IT initiatives. Start by establishing the IT department as the social authority expert in the field of green IT and that the benefits that can be derived with minimal disruption to the workforce.
By becoming the social authority, you become the influencer and have the opportunity to pull other employees outside the IT department into the push towards green IT. One of the foundational concepts of social media, however, is that you cannot completely control your message through social media but you begin to participate in the conversation with the objective of becoming a relevant influence in it. Be careful in your conversations, as the participation must be cleverly executed; people are resistant to direct or overt 'marketing'. Credibility is important: Always back up your statements with clear facts, but do not overwhelm conversation participants. Build trust slowly but continually. The most effective marketing approaches revolve around honestly convincing people of genuine intention, knowledge and expertise in green IT through providing valuable and accurate information on an on-going basis with a marketing angle overtly associated. If this approach is successfully executed, the message itself will begin to develop naturally, the IT organization will be viewed as the authority and users will naturally begin to gravitate to them for additional information. No project can be successful without comprehensive communication to all levels within the company. Everyone must support the move to green IT, but in order to support it, they must know about it and understand the value.

These tools help spread awareness, engage stakeholders, and drive action toward environmentally sustainable IT practices.

1. Raising Awareness about Green IT

Role of Communication:

Clear communication channels within organizations ensure employees, management, and stakeholders understand the importance and benefits of Green IT.

Example: Internal newsletters, webinars, and training sessions to explain energy-saving practices.

Role of Social Media:

Platforms like LinkedIn, Twitter, and Facebook can spread Green IT awareness on a global scale.

Example: Companies sharing case studies and success stories about their Green IT projects to inspire others.

2. Engaging Stakeholders

Role of Communication:

Regular updates about Green IT initiatives, goals, and progress keep stakeholders informed and invested in the process.

Example: Monthly reports or presentations about reductions in energy consumption or carbon footprint.

Role of Social Media:

Social media platforms enable direct interaction with stakeholders, such as customers, partners, and employees.

Example: Hosting live Q&A sessions or polls to discuss Green IT strategies and gather feedback.

3. Promoting Green IT Practices

Role of Communication:

Internal communication helps embed Green IT principles in daily operations. This includes educating employees about e-waste recycling, energy-efficient device usage, and cloud computing.

Example: Sharing guidelines for reducing power usage on personal devices.

Role of Social Media:

Social media campaigns can encourage mass adoption of sustainable IT practices.

Example: Running a campaign on Earth Day to promote using energy-efficient technologies.

4. Driving Behavioral Change

Role of Communication:

Clear messaging helps foster a culture of sustainability within organizations, motivating employees to adopt eco-friendly IT practices.

Example: Emails or posters promoting small but impactful actions, such as turning off unused devices.

Role of Social Media:

Social media platforms create a sense of community and accountability by showcasing collective efforts.

Example: Encouraging the use of hashtags like #GreenIT or #SustainableTech to share eco-friendly tips and accomplishments.

5. Advocating for Policy and Regulation

Role of Communication:

Effective communication with policymakers and regulatory bodies can influence the development of Green IT-related laws and standards.

Example: Industry conferences or reports highlighting the need for energy-efficient IT regulations.

Role of Social Media:

Social platforms amplify advocacy efforts by reaching wider audiences, creating public pressure for policy change.

Example: Viral campaigns urging governments to promote renewable energy in data centers.

6. Monitoring and Sharing Progress

Role of Communication:

Transparent communication of Green IT goals and achievements builds trust and accountability.

Example: Publishing sustainability reports highlighting progress in reducing carbon emissions.

Role of Social Media:

Social media is an excellent platform for showcasing milestones and benchmarking against peers.

Example: Sharing infographics on LinkedIn about reduced energy consumption after implementing virtualization.

7. Building Partnerships

Role of Communication:

Collaborating with partners, vendors, and NGOs through effective communication ensures alignment in Green IT goals.

Example: Partnering with recycling firms to manage e-waste.

Role of Social Media:

Social platforms connect organizations with like-minded partners and foster collaborative efforts.

Example: Joining or creating online communities to exchange best practices for sustainable IT.

Examples of Social Media in Green IT

IBM's Green IT Campaigns: IBM uses Twitter and LinkedIn to promote its energy-efficient technologies and sustainability goals.

Microsoft Cloud for Sustainability: Microsoft shares updates on its cloud solutions that help businesses track and reduce their carbon footprints.

Q.6

A Explain Green Cloud computing Architecture.

Green Cloud Computing Architecture refers to the design and implementation of cloud computing systems that aim to reduce energy consumption and minimize environmental impact. This architecture focuses on leveraging energy-efficient technologies, renewable energy sources, and sustainable operational practices while maintaining performance and scalability.

In the green cloud architecture, users submit their cloud service requests through a new middleware Green Broker that manages the selection of the greenest cloud provider to serve the user's request. A user service request can be of three types: software, platform or infrastructure.

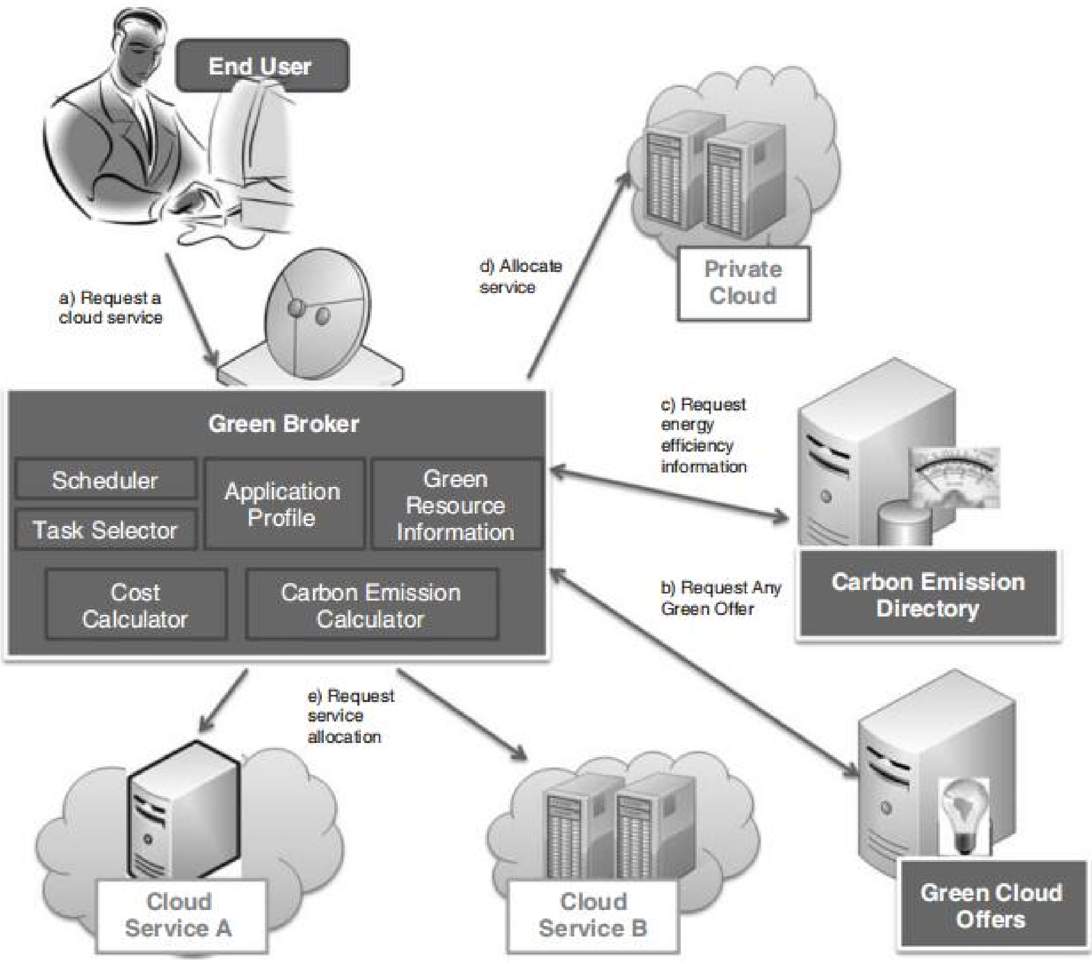

Figure 16.6 Green cloud architecture.

The cloud providers can register their services in the form of 'green offers' to a public directory which is accessed by Green Broker. The green offers consist of green services, pricing and time when it should be accessed for least carbon emission. Green Broker gets the current status of energy parameters for using various cloud services from the Carbon Emission Directory. The Carbon Emission Directory maintains all the data related to cloud service energy efficiency. These data may include the PUE and cooling efficiency of the cloud data centre which is providing the service, network cost and carbon emission rate of electricity. Green Broker calculates the carbon emission of all the cloud providers who are offering the requested cloud service. Then, it selects the set of services that will result in the least carbon emissions and buy these services on behalf of users.

The green cloud framework is designed such that it keeps track of the overall energy usage of serving a user request. It relies on two main components, the Carbon Emission Directory and green cloud offers, which keep track of the energy efficiency of each cloud provider and also give incentive to cloud providers to make their service 'green'. From the user side, Green Broker plays a crucial role in monitoring and selecting the cloud services based on the user QoS requirements, and ensuring minimum carbon emissions (MCEs) for serving a user. In general, a user can use cloud to access any of these three types of services (SaaS, PaaS and IaaS), and therefore the process of serving them should also be energy efficient. In other words, from the cloud provider side, each cloud layer needs to be 'green' conscious.

- **SaaS level:** Since SaaS providers mainly offer software installed on their own data centres or resources from IaaS providers, the SaaS providers need to model and measure the energy efficiency of their software design, implementation and deployment. For serving users, the SaaS provider chooses the data centres which are not only energy efficient but also near to users. The minimum number of replicas of users' confidential data should be maintained using energy-efficient storage.

- **PaaS level:** PaaS providers offer in general the platform services for application development. The platform facilitates the development of applications which ensure system-wide energy efficiency. This can be done by including various energy-profiling tools such as JouleSort (Rivoire *et al*., 2007), a software energy efficiency benchmark that measures the energy required to perform an external sort. In addition, platforms can be designed to have various code-level optimizations which can cooperate with underlying compliers in the energy-efficient execution of applications. Other than application development, cloud platforms also allow the deployment of user applications on the hybrid cloud. In this case, to achieve maximum energy efficiency, the platforms profile the application and decide which portion of the application or data should be processed in-house and in the cloud.

• **IaaS level:** Providers in this layer play the most crucial role in the success of the whole green architecture since the IaaS level not only offers independent infrastructure services but also supports other services offered by clouds. They use the latest technologies for IT and cooling systems to have the most energy-efficient infrastructure. By using virtualization and consolidation, energy consumption is further reduced by switching off unutilized servers. Various energy meters and sensors are installed to calculate the current energy efficiency of each IaaS provider and its sites. This information is advertised regularly by cloud providers in the Carbon Emission Directory. Various green-scheduling and resource-provisioning policies will ensure minimum energy usage. In addition, the cloud provider designs various green offers and pricing schemes for providing incentives to users to use their services during off-peak or maximum energy efficiency hours.

B Explain processor power state in detail.

3.1.1 Processor Power States
For software developers, the component of primary interest is the central processing unit (CPU). CPUs have defined energy states as well. It makes sense that if the CPU is not actively processing information or performing computations, it should be consuming minimal energy. The CPU has what are called C-states and P-states (Intel, 2011). *C-states* are core power states that define the degree to which the processor is 'sleeping'. In state C0, the processor is active and executing instructions. Whilst in C0, the processor can operate at various frequency levels, designated as *P-states* (performance states).

3.1.1.1 C-States
In the C0 state the CPU is active – it is busy carrying out some task, and it is performing that task at a frequency (P-state) that is appropriate. Between periods of activity the CPU can take the opportunity to rest or 'sleep'. In fact, C-states are often referred to as sleep states. Intel processors support several levels of core and package (resources shared by all the cores) C-states that provide a flexible selection between power consumption and responsiveness. With each successively deeper sleep state, some new part of the CPU is turned off and more energy is saved. Deeper sleep leads to greater energy savings. Even if the sleep period is only 100 μs, a considerable amount of energy can be saved over time.
Common C-States:

C0 (Active State):
This is the full-power state where the processor is actively executing instructions.
It consumes the most power because all processor components are running at full speed.

C1 (Idle State):
The processor is still powered on, but it reduces its clock speed and halts certain parts of the core to save power.
This state offers minimal power savings but allows for a very fast return to C0 when the processor is required to process tasks.

C2 (Sleep State):
In this state, more parts of the processor are powered down, resulting in greater power savings compared to C1.
The return to the active state (C0) from C2 is slightly slower than C1 but still relatively fast.

C3 (Deep Sleep State):
Almost all processor components are turned off, except for some circuits that monitor for interrupt requests or tasks.
This state offers significant power savings but has a longer wake-up time compared to C0, C1, and C2.

C4 and Beyond (Deeper Sleep States):
These are even deeper idle states where the processor consumes very little power.
The wake-up time becomes increasingly longer as you move to higher C-states, but power savings are maximized.

3.1.1.2 P-States
P-states define the frequency at which the processor is running. Different processor brands showcase their P-states as features, such as SpeedStep in Intel processors, PowerNow! or Cool'n'Quiet in Advanced Micro Devices (AMD) processors and PowerSaver in VIA processors. Typical P-states are as follows:
• P0; maximum power and frequency.
• P1; less than P0; voltage or frequency scaled down.
• Pn; lowest rated voltage and/or frequency.
P-states were developed to save energy overall and deliver the performance you need when you need it. We can see how P-states save energy by studying the following equation:
$P = CV2f$
This means that the power (P) required to run a CPU is the product of the capacitance (C), the frequency (f) and the voltage V. Assuming constant capacitance and voltage, reducing the frequency leads to a corresponding reduction in power. In practice, P-states are realized by adjustments to both voltage and frequency. In general, we consider a lower P-state to be a lower frequency state. An end user can actually control (to a moderate extent) the P-state of the processor by setting the platform power policy

Common P-States:

P0 (Maximum Performance):

The processor runs at its maximum frequency and voltage, providing the highest performance for demanding tasks.

This state consumes the most power.

P1, P2, P3, etc. (Reduced Performance States):

These states represent various levels of reduced performance, with each level decreasing the processor's frequency and voltage.

As the P-state number increases, both power consumption and performance decrease.

Pn (Minimum or Idle State):

The processor is in the lowest performance state, where it runs at a minimal frequency, significantly lowering power consumption.

The processor enters Pn during periods of inactivity or light tasks.

Write short notes on:

a) Nine principles of green engineering

A new discipline called green engineering is taking shape now and receiving interest. Green engineering instills environmentally conscious attitudes, values and principles, combined with science, technology and engineering practice, all directed towards improving local and global environmental quality (Virginia Tech, 2012). It promotes the adoption of environmentally friendly principles and practices early in the design and development phase of a process or product. It focusses on the design of materials, processes, systems and devices with the objective of minimizing overall environmental impact (including energy utilization and waste generation and management) throughout the entire life cycle of a product or process – from initial extraction of raw materials used in manufacture to ultimate disposal at the end of the useful life of a product.

Green engineering uses the following nine principles as guidance in the design or redesign of products and processes satisfying the constraints imposed by business, government and society such as cost, safety, performance and environmental impact

1. Engineer processes and products holistically, carry out systems analysis and integrate environmental impact assessment tools.

2. Conserve and improve natural ecosystems whilst protecting human health and well being.

3. Use life cycle thinking and considerations in all stages of activities.

4. Ensure that all material and energy inputs and outputs are inherently environmentally safe.

5. Minimize the depletion of natural resources.

6. Strive to prevent waste.

7. Develop and implement solutions and be cognizant of local geography, aspirations and cultures.

8. Create solutions beyond current or dominant technologies; improve, innovate and invent (technologies) to achieve sustainability.

9. Actively engage communities and stakeholders in the development of products and processes and in the entire product life cycle.

b) Any 2 hazardous chemicals used in the manufacturing of electronic devices

• **Lead** is generally used in printed circuit boards and in glass monitors. The pathological effects of lead are most prominent in the nervous system, hematopoietic system and kidneys. Excessive exposure to lead (Pb) results in clinical toxicity. There has been extensive research to find a replacement for lead (Norwegian University of Science and Technology, 2010). There are also on-going efforts to recover lead from the e-waste

• **Cadmium** is generally used in low-temperature soldering, plating for corrosion protection, colorants in plastics and contact buttons in relays Cadmium when ingested affects the liver and kidneys. Research is on-going to produce alternatives for its various uses.

• **Mercury** is mainly used in monitors and batteries. Mercury is a well-known pollutant , and it affects the immune system, alters genetic and enzyme systems and damages the nervous system. New light-emitting diode (LED) backlit technologies have been developed which make monitors mercury free. Mercury use in batteries has also been reduced considerably over the past few years.

• **PBDEs and PBBs** are used as flame retardants (Wisconsin Department of Natural Resources, 2011). These materials are ubiquitous environmental pollutants; they are extremely toxic and are banned in many countries. Efforts are on-going to remove these materials from the manufacture of computers.

• **Arsenic** is used in glasses and in manufacturing semiconductors. Arsenic interferes with cellular longevity by allosteric inhibition of an essential metabolic enzyme. Many arsenic prevention and recovery methodologies have been devised, and these need to be actively deployed to reduce arsenic usage in the manufacturing process.

• **PVC** is one of the most commonly used plastics worldwide. It has been used extensively in the manufacture of computer parts. Vinyl chloride, one of the main ingredients of PVC, is a carcinogen and also affects the human reproductive system . Of late, many companies have eliminated the use of PVC in their devices.

c) Blade servers

The blade system design differs from that of a rack design in some fundamental ways. Racked servers are self-contained individual machines with power and network cabling for each unit. In contrast, blades are housed within a blade system container, which is mounted in a standard cabinet. The blade system provides the power and networking for all the blades housed within the system. Onboard cooling and uninterruptible power supply (UPS) requirements may also be provided by the blade cabinet system. This design, whereby the common support infrastructure (power, networking and cooling) is separated out into a common house system, enables greater density of machine within a standard cabinet – 42 servers for a rack scales to 128 for blades, giving a far greater density or capacity using the blade system design.
Physical dimension may at first not appear relevant to green data centres. But as racks transition to blades, the capacity of the data centre in terms of hosting physical machines also increases as does the energy requirement.

d) Green Enterprise

A Green Enterprise is a business organization that incorporates environmentally sustainable practices into its operations, strategies, and goals. The primary objective of a Green Enterprise is to minimize its ecological footprint while contributing positively to environmental sustainability. This includes adopting eco-friendly policies, using sustainable resources, reducing waste, and leveraging energy-efficient technologies.

Key Characteristics of a Green Enterprise

Sustainable Resource Management:
Ensures the efficient use of natural resources such as water, energy, and raw materials.
Promotes the use of renewable resources to reduce dependence on non-renewable ones.

Energy Efficiency:
Implements practices and technologies that reduce energy consumption across all operations.
Incorporates energy-efficient lighting, HVAC systems, and machinery.

Waste Reduction and Recycling:

Establishes processes to reduce, reuse, and recycle waste products.
Properly manages hazardous and electronic waste to minimize environmental harm.

Adoption of Green IT:
Uses energy-efficient servers, virtualization, cloud computing, and other Green IT technologies to reduce energy use in IT infrastructure.

Carbon Footprint Reduction:
Monitors and reduces greenhouse gas emissions through cleaner production processes and renewable energy sources.
Implements carbon offset initiatives to compensate for unavoidable emissions.

Eco-Friendly Products and Services:
Designs products that are energy-efficient, durable, and recyclable.
Offers services that align with sustainability goals, such as paperless billing or green logistics.

Employee Engagement:
Involves employees in green initiatives through training, awareness programs, and incentives.
Encourages eco-friendly practices in the workplace, such as carpooling, remote working, or reducing paper use.

Sustainable Supply Chain:
Works with suppliers who follow environmentally friendly practices.
Reduces the environmental impact of transportation and packaging

Examples of Green Enterprise Practices
Technology Companies:
Companies like Google and Microsoft use renewable energy to power data centers and are committed to carbon neutrality.

Retail and Manufacturing:
IKEA focuses on using sustainable raw materials and has invested in renewable energy projects.

Financial Services:
Banks like HSBC promote green financing by investing in eco-friendly projects and offering green bonds.

Q.2

(b) Explain recycle, reuse and refurbish in green computing in detail and with an example

1.5.2 The Three Rs of Green IT

Unwanted computers, monitors and other hardware should not be thrown away as rubbish, as they will then end up in landfills and cause serious environmental problems. Instead, we should refurbish and reuse them, or dispose them in environmentally sound ways.

Reuse, *refurbish* and *recycle* are the three 'Rs' of greening unwanted hardware.

• **Reuse.** Many organizations and individuals buy new computers for each project or once every 2–3 years. Instead, we should make use of an older computer if it meets our requirements. Otherwise, we should give it to someone who could use it in another project or unit. By using hardware for a longer period of time, we can reduce the total environmental footprint caused by computer manufacturing and disposal.

Example:

- An organization upgrades its office computers. The older computers, still in working condition, are donated to schools or community centers where they can be used for educational purposes.

• **Refurbish.** We can refurbish and upgrade old computers and servers to meet our new requirements. We can make an old computer and other IT hardware almost new again by reconditioning and replacing some parts. Rather than buying a new computer to our specifications, we can also buy refurbished IT hardware in the market. More enterprises are now open to purchasing refurbished IT hardware, and the market for refurbished equipment is growing. If these options are unsuitable, we can donate the equipment to charities, schools or someone in need, or we can trade in our computers.

Example:

- A used smartphone with a broken screen and outdated software is refurbished. The screen is replaced, the operating system is updated, and the phone is resold as a certified refurbished device at a lower cost.

• **Recycle.** When we cannot refurbish or otherwise reuse computers, we must dispose of them in environmentally friendly ways by depositing them with recognized electronic recyclers or electronic waste (e-waste) collectors. E-waste – discarded computers and electronic goods – is one of the fastest-growing waste types and poses serious environmental problems. The United Nations Environment Program estimates that 20–50 million tons of e-waste are generated worldwide each year, and this is increasing. IT hardware contains toxic materials like lead, chromium, cadmium and mercury. If we bury IT hardware in landfills, toxic materials can leach harmful chemicals into water ways and the environment. If burned, they release toxic gases into the air we breathe. So if e-waste is not discarded properly, it can harm the environment and us. Waste electrical and electronic equipment (WEEE) regulations aim to reduce the amount of e-waste going to landfills and increase recovery and recycling rates.

Example:

- An old laptop is dismantled at a recycling facility. The circuit board is processed to recover gold and copper, the battery is recycled for its lithium, and the plastic casing is used to create new plastic products.

Disposal of e-waste, if not properly done, causes serious environmental damage and health problems particularly to those directly involved in the disposal or recycling. Despite bans on the export and import of e-waste, e-waste gets into developing countries (such as India, China and Philippines) for ‘recycling’ as the cost of recycling is lower there. Unfortunately, as environmental regulations and proper means of e-waste disposal and recycling are not enforced in practice in these countries, e-waste is handled ‘informally’ in unofficial recycling markets by manual, crude, hazardous means to extract metals and other valuables. Computer manufacturers should take their share of responsibility and take action to reduce pollution caused by their products’ end of life.

Q.3

(b) Explain storage media power characteristics

The operational characteristics of different devices highly influence their power consumption. In this section, firstly we describe the different operational states and their power characteristics in hard disks, followed by the characteristics of tapes and conclude by summarizing the characteristics of an emerging storage media type, SSDs.

6.2.1 Hard Disks

Hard disks are the most common nonvolatile storage media. A hard disk drive (HDD) contains disk platters on a rotating spindle and read-write heads floating above the platters. The read-write heads encode data magnetically. Power consumption of a hard disk in different operational (power) states can differ considerably, and energy management techniques aim to effectively exploit this feature. Power consumption of magnetic hard disks is a function of its rotational speed and the data access rate. Most power is consumed by the rotating spindle, followed by the head assembly that moves along the platters to the requested sectors or logical block addresses (LBAs) and the buffers used for queuing requests and requested data. The total power consumption (*Ptotal*) of a hard disk is the sum of the power consumed by the spindle motor (*Pspindle*), the power consumed by the head movement (*Phead*) and the power consumed by other components (*Pother*) which is relatively small, has less variation and includes the power consumed by buffers (*Pbuffer*).

Ptotal = Pspindle + Phead + Pother

The power consumed by the spindle motor (*Pspindle*) is directly proportional to the square of its angular velocity (ω), that is, , $P_{spindle} \propto \omega^2$. . A hard disk is either rotating at its full speed or not rotating at all; therefore, *Pspindle is power consumed at full speed (Pfullspeed) or zero*. The power consumed by the head movement (*Phead*) is dictated by the disk access pattern. Random access causes more head movements than sequential accesses, and thus leads to higher head power consumption

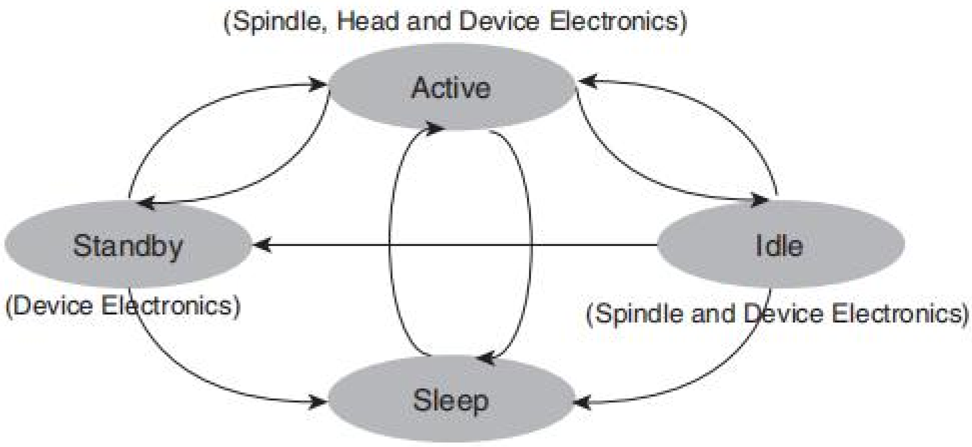

Figure 6.2 States of mechanical hard disk drives.

6.2.2 Magnetic Tapes

Magnetic tape is another medium for data storage. It is made of a thin magnetic coating on a long, narrow strip of plastic. To store data on magnetic tapes, a tape drive uses a motor to wind the magnetic tape from one end to another and through tape heads to read, write and erase the data. Magnetic tape is still a major data storage medium especially in some industries such as radio and TV broadcasting because of its very low storage-per-bit cost compared to that of hard disks. Although its areal density is lower than the density of disks, the capacity of tape is still very high (i.e. similar to disks) due to its larger amount of surface area compared to disk surface. The large capacity and cost advantage makes magnetic tapes a viable alternative, particularly for backup and archiving. Further, the average energy consumption of tape drive is much less compared to a hard drive as it consumes power only when it reads or writes. Power consumption in the idle or retention state is zero. The cost of maintaining a tape library, which can be amortized over thousands of tape cartridges, is negligible.

6.2.3 Solid-State Drives (SSDs)

Recently, SSDs have gained increased adoption due to their better performance potential and higher power efficiency. Unlike electromechanical HDDs which contain spinning disks and movable heads, a SSD uses solid-state memory (i.e. nonvolatile memory) to store persistent data and has no moving parts. Hence, compared to HDDs, SSDs have lower access time and power consumption.

Today's SSDs use nonvolatile *not AND* (NAND) flash memory. There are two types of flash memory: multilevel cell (MLC) and single-level cell (SLC) flash memory. SLCs store a single bit in a single memory cell, whereas MLCs store multiple bits in a single cell by allowing each cell to store multiple electrical states. For example,a four-level MLC stores 2 bits in a single memory cell. Typically, MLCs are cheaper and have greater storage density, but are slower than SLCs. SSD has rapidly increased in popularity as the primary data storage medium for mobile devices, such as phones, digital cameras and sensor devices, notebook computers and tablets. Its features include small size, low weight, low power consumption, high shock resistance and fast read performance. SSD has also started to penetrate the markets from laptops and PCs to enterprise-class server domains. Enterprise-class SSDs provide unique opportunities to boost I/O-intensive applications' performance in data centres. Compared to hard disks, SSDs are very attractive for the high-end servers of data centres due to their faster read performance, lower cooling cost and higher power savings. SSDs have the lowest power consumption rate of active devices when compared with both dynamic random-access memory (DRAM) and hard disks. For example, the power consumption for a 128 GB SSD is about 2 W, that of a 1 GB DRAM dual in-line memory module (DIMM) module is 5 W, and for a 15 000 RPM, 300 GB (7200 RPM, 750 GB) hard drive, it is around 17.5 W (12.6 W). This efficiency can mainly be attributed to the nonmechanical nature of these devices. Data in SSD is read or written in units of pages. The read time for a page in SSD is about 20 times faster than a page read from hard disks. Unlike hard disks, there are no differences in terms of time between random reads and sequential reads. SSD is best suited for caching data between memory and hard disks since the hard disks still hold the advantage of cost and storage capacity. However, the relatively low write performance and longevity problems of SSD require new and innovative solutions to fully incorporate SSDs into the high-end servers of data centres. Unlike conventional disks, where read and write operations exhibit symmetric speed, in non-enterprise SSDs, the write operation is substantially slower than the read operation. This asymmetry arises because flash memory does not allow overwrite and write operations in a flash memory must be preceded by an erase operation to a block. Typically a block spans 64–128 pages, and live pages from a block need to be moved to new pages before the erase can be done. Furthermore, a flash memory block can be erased only a limited number of times. The slow write performance of non-enterprise SSDs and the wear-out issues are the two major concerns. Existing approaches to overcome these problems through modified flash translation layer (FTL) are effective for sequential write patterns; however, random write patterns are still very challenging.

Q.4

(b) Explain server farm and grid computing

5.2.4.1 Server Farm (Cluster)

Cluster computing is characterized by multiple, physically discrete machines, closely linked to provide the logical interface of a single machine. Often associated with the parallelization of processing algorithms, cluster computing requires dedicated and highly specialized middleware to form the complex message-passing infrastructure required to manage the cluster's physical resources. The parallel processing capability would split and schedule individual requests across the large number of individual server machines, each configured to compute a result. The results of individual processes would then be collated together to form the final result. The savings in parallelizing the workload is the principle use and research interest for cluster computing.

5.2.4.2 Grid Computing

The increasing demands of the scientific research community gave rise to research into grid computing as a natural progression to clusters. As data sets increased in size and the demands for processing capabilities to analyse data in a timely manner also grew, a new approach was sought. The core aim of grid computing was to integrate disparate resources across organizational domains into what became termed virtual organizations. For example, a database server in one domain may be integrated with a networked cluster in another domain, to form a powerful data analytics platform. Grid computing, however, presents a few issues including security management across multiple domains, the complexity of handling the interaction state and the realities of processing vast data sets across a network. As a result the approach did not gain traction beyond academic circles. It was not until some clever thinking and the adoption of a Web services approach – common in the e-commerce domain – that a breakthrough was made in Internet-scale data analytics.

Q.5

(a) What is state transitioning and caching for hard disks?

To reduce the energy consumption of hard disks, different techniques and methodologies are being adopted. Three commonly used energy management techniques for hard disks are state transitioning, caching and dynamic RPM.

6.3.1 State Transitioning

Given that in a hard disk, the spindle motor consumes most of the power, state transitioning techniques try to turn off the spindle motor or keep it in standby mode during idle periods. The disk transitions to standby or off mode if there is no request to be served. If the disk idle period is not long enough, the time overhead of *spin down* and *spin up* can affect the disk response time significantly. Moreover, the power consumed by transitioning itself might exceed the power saving gained from a low-power state during the short idle period. If the disk is already idled for the threshold time, it transitions to standby mode. If it stays in standby mode for another threshold of time without requests, it can further transition to off mode. In this approach, the historical information is used to predict the future access pattern. Different variations on access prediction are based on historical information. Most on-going research and development in state transitioning revolve around idle period prediction and minimizing the performance impact of these transitions on disk responsiveness (as the transition time is usually around 8–10 seconds). Some state-transitioning techniques provide a performance guarantee

6.3.2 Caching

In order to speed up access for both read and write requests, enterprise storage solutions typically have huge amounts of cache in conjunction with regular disks. Further, to make use of the cache to aid in disk power management, various techniques are recommended. These cache management techniques or algorithms aim to minimize disk power usage, either by minimizing disk access or by increasing the length of idle periods. One could in effect use huge caches to increase the idle periods of disks and in doing so can help more disks to transition to the sleep state, thereby improving energy efficiency. The cache management algorithms partition-aware least recently used and partition-based LRU are centred on this idea. PALRU classifies all disks based on access patterns into two classes – priority (disks with fewer cold misses and longer idle times) and regular – and maintains two separate LRU queues. At the time of an eviction decision, first the regular queue elements are chosen as victims. If the regular queue is empty, the algorithm chooses elements from the priority queue. PBLRU, however, differentiates between disks by dynamically varying the number of allocated cache blocks per disk. It divides the cache into multiple partitions (one per disk) and adjusts the size of these partitions peri odically based on workload characteristics. The simulation results with online transaction processing (OLTP) traces show that PALRU consumes 14–16% less energy and PBLRU consumes 11–13% less energy than traditional LRU. On the other hand, for the Cello96 trace (file system trace), PALRU saves less than 1% energy over LRU whilst PBLRU is 7.6–7.7% more energy efficient than LRU. Since write requests in enterprise storage devices almost never get written directly to target disks (they are cached instead), another technique is to use write offloading as a mechanism to conserve disk power usage . Write offloading facilitates complete spin downs of volumes periodically, thereby aiding in significant power savings. By using write offloading, about 45–60% energy savings can be achieved in write-dominated application environments.

(b) Explain the four objectives of energy optimization

Four Objectives of Energy Optimization

Energy optimization refers to the process of improving energy efficiency to minimize energy consumption, reduce costs, and achieve environmental sustainability. In green computing and broader industrial applications, the key objectives of energy optimization include:

1. Reduce Energy Consumption

The primary goal of energy optimization is to reduce the amount of energy used by systems, devices, and operations without compromising performance or functionality.

Strategies:

Implement energy-efficient hardware (e.g., low-power processors, LED lighting).

Use software solutions that enable dynamic power management, such as reducing power in idle states.

Optimize data center cooling systems with advanced technologies like liquid cooling.

Benefits:

Decreases operational costs by lowering energy bills.

Conserves natural resources used to generate electricity.

Reduces the carbon footprint associated with energy production.

Example: Virtualizing servers in a data center reduces the number of physical machines required, significantly lowering energy consumption.

2. Maximize System Efficiency

Energy optimization focuses on maximizing the efficiency of systems by ensuring that every watt of energy consumed contributes effectively to achieving desired outputs.

Strategies:

Optimize workloads to balance energy use and performance (e.g., load balancing in cloud computing).

Use machine learning algorithms to predict and adjust energy requirements dynamically.

Schedule non-critical tasks during off-peak hours to avoid energy wastage.

Benefits:

Improves the overall performance-to-energy ratio.

Enhances the lifespan of hardware by reducing unnecessary energy stress.

Helps achieve higher productivity with minimal energy input.
Example: In manufacturing, adjusting conveyor belt speeds based on workload ensures that energy is used efficiently without overburdening the system.

3. Reduce Environmental Impact
Energy optimization aims to minimize the environmental harm caused by energy use, such as greenhouse gas emissions, resource depletion, and waste generation.

Strategies:
Transition to renewable energy sources like solar, wind, or hydropower to power operations.
Reduce dependency on fossil fuels by adopting green technologies.
Use energy-efficient building designs (e.g., smart buildings with energy monitoring systems).

Benefits:
Helps mitigate climate change by reducing carbon emissions.
Conserves ecosystems by minimizing pollution and resource extraction.
Supports corporate social responsibility goals.
Example: Google powers its data centers with renewable energy, significantly lowering its environmental footprint while maintaining energy-intensive operations.

4. Achieve Cost Savings
Energy optimization also focuses on reducing costs by improving energy use efficiency and adopting cost-effective energy sources.

Strategies:
Use predictive analytics to identify areas of high energy waste and optimize them.
Implement energy-saving technologies such as LED lighting and smart meters.
Negotiate with energy providers for better rates or invest in on-site renewable energy generation.

Benefits:
Lower energy bills directly reduce operational expenses.
Minimizes the need for additional investments in infrastructure due to reduced energy demands.
Improves profitability by reallocating saved costs to other critical business areas.

Example: A company installing solar panels on its premises may incur upfront costs but saves significantly on energy bills over time, eventually achieving cost neutrality and even profit.

Conclusion

The four objectives of energy optimization—reducing energy consumption, maximizing system efficiency, reducing environmental impact, and achieving cost savings—work together to promote sustainable energy use. These objectives not only support environmental goals but also align with business objectives, making energy optimization a critical strategy for modern organizations striving for sustainability and profitability

Q.6

(a) Explain the business drivers of green IT strategy

Businesses need compelling reasons to undertake and implement green IT strategies. Business drivers of green IT (Trivedi and Unhelkar, 2010; Unhelkar, 2011) can be grouped into six categories as shown in Figure 8.3: (i) costs (including energy costs and operational costs), (ii) regulatory and legal, (iii) sociocultural and political, (iv) new market oppor tunities, (v) enlightened self-interest and (vi) a responsible business eco-system. Green IT strategies, policies, design, implementation and practice find necessary impetus by a combination of one or more of these drivers. The influence of these drivers on green IT strategies are discussed in this section.

8.3.1 Cost Reduction

Cost reductions provide an excellent driver for an organization to come up with a comprehensive green IT strategy. As a result of a green initiative, cost reduction could be Business drivers for green IT strategies). derived from minimizing energy consumption (improving energy efficiency), reducing the use of raw materials and equipment, recycling equipment and waste and optimizing storage and inventory. Whilst efforts to reduce costs can provide an impetus for carbon emission reduction, organizations undertaking green transformations need to be aware of the investment that they must incur as a result of their greening effort. For example, optimizing a business process can eliminate the need for a desktop machine but, instead, there may be a need to replace that desktop with a mobile device. Virtualizing a data centre, whilst improving resource utilization and reducing cooling costs, will require some initial investment from the business for implementing virtualization. At the organizational level, costs associated with a green enterprise transformation programme need to be factored in along with the anticipated cost reduction due to the transformation.

8.3.2 Demands from Legal and Regulatory Requirements

Government rules and regulations comprise a major driver for many green enterprise transformation programmes. The relative importance given to the regulatory factor, as compared with other factors such as organization self-initiation, customer demand and pressure from society, are the highest – 70% as reported by Unhelkar (2011). Regulatory acts such as National Greenhouse and Energy Reporting (NGER) (www.climatechange.gov.au/reporting; Australian Government, 2011a) and the Carbon Pollution Reduction Scheme (CPRS) require organizations to mandatorily report their carbon emissions if they are above a certain threshold level. Regulatory bodies also provide some basic calculators to enable greenhouse gas calculations. An example one such calculator is the NGER calculator OSCAR (Online System for Comprehensive Activity Reporting; Australian Government, 2011b). These calculators are used to determine an organization's total carbon emissions that can be used to decide whether the organization will require mandatory reporting. In addition to the basic calculators, green information systems also source external regulatory data (such as permissible emissions figures), and store, analyse and broadcast the results that enable an organization to monitor and improve its performance. These organization-specific green information systems are more sophisticated than the basic calculators provided by the regulatory bodies (Unhelkar and Philipson, 2009).

8.3.3 Sociocultural and Political Pressure

Sociocultural and political pressure become major driving forces when an organization's society recognizes the environment as of significant value and is interested in protecting it. Such acceptance of the environment's importance by the society brings pressure on the organization to change. For example, the increasing popularity of and adherence to Earth Hour (last Saturday of March in most countries), wherein almost all large edifices around the world switch off their electrical power for all non-essential things for an hour, or Earth Day (22 April in the United States and 20 March by the United Nations) have a corre sponding bearing on many large businesses' sustainability strategies. This 'groundswell' of opinions also leads to corresponding shifts in political viewpoint. As a result, the organization is forced to seriously reconsider its business priorities and processes in light of the environment. Whilst the scale and nature of the benefits of such CSR for an organization can vary depending on the nature of the enterprise, and are difficult to quantify (Garito, 2011), their importance should not be discounted.

8.3.4 Enlightened Self-Interest

Self-interest comes into play when an organization, on its own accord, realizes the need to be, and the benefits of being, environmentally responsible and creates or adopts a green strategy. It may include a range of interests including the organization's desire to undertake a genuine *common* good, the need of business leadership to achieve personal satisfaction or maintain or raise employee morale or simply the decision makers' understanding that costs can be reduced and customers more satisfied with a self-interest approach that also helps the environment. The desire to achieve brand recognition based around environmental sustainability or an understanding of its impact on business continuity also forms part of this driver (Cartland, 2005). This driver can thus translate into self-motivation and has the potential to be an effective green driver for businesses. A variation of this driver, known as incentive-driven compliance (IDC), incorporates innovation and self-motivation within its environmental approach for better carbon compliance.

8.3.5 Collaborative Business Ecosystem

If a large organization that has myriad different associations with its many collaborating smaller sized organizations changes its direction and priorities, then those collaborating organizations also have to change their priorities accordingly. When such a large organization embarks on environmentally sustainability programmes in a major way encompassing its supply chain, an entire ecosystem made up of the business partners, suppliers and customers and internal users organizations, together with the industry and the corresponding business consortiums in which the organization exists, is affected. These various stakeholders and associations are invariably pushed into implementing environ mentally responsible initiatives and strategies. This happens by virtue of the multiple interactions – physical and electronic – that are undertaken in the course of daily business activities. This scenario is demonstrated by HP, wherein not only are the environmental impacts monitored and managed by the organization, but also, by virtue of its own management and active involvement with the members of its supply chain, the overall carbon impact of the activities of HP's suppliers is also reduced (Velte, Velte and Elsenpeter, 2008). In addition to the impact of collaborating organizations in a business ecosystem, there are also considerations of superimposition of long-term trends upon short-term markets (Goel *et al*., 2011). Such superimpositions bring about major business changes and restructuring in favour of ideas that are environmentally friendly.

8.3.6 New Market Opportunities

Global environmental awareness, corresponding legislations and sociocultural and political pressure on businesses have created opportunities for new markets that did exist or were not even envisaged a few years ago. For instance, these new markets can create and provide products and services that assist other organizations in achieving their green initiatives and goals. Thus, we are talking about not only 'businesses that are green' but also 'green as a business offering'. For example, carbon emissions management software (CEMS) is a new breed of software applications that are now available. The developers of these new software applications have discovered a market that did not exist before. Similarly, smart meters to measure carbon emissions, opportunities to apply new standards for the optimization of emissions and new architecture and design of low-carbon gadgets comprise markets that are likely to grow in the carbon economy. Despite the discussion on the aforementioned drivers for businesses to undertake green initiatives, practical experience suggests that these green strategy drivers are usually interpreted by organizations in their own ways. For example, as described by Godbole (2011), within the health care sector, hospitals have significant challenges with disposal of hazardous waste, whilst insurance companies are more concerned with reducing paper use or decreasing power consumption in their data centres. Thus, in practice, these drivers will result in a combination of drivers for the business to initiate green IT – depending on what it considers as its own key environmental and business issues. In the development of a green IT strategy, not only do these drivers need independent analysis, but also they need to be studied together to see their overall impact on the organization.

(b) Explain the cloud computing deployment models

Cloud computing is an evolving paradigm which enables the delivery of IT needs such as storage, computation and software such as office and enterprise resource planning (ERP) through the Internet. The shift towards such service-oriented computing is driven primarily by ease of management and administration processes involving software upgrades and bug fixes. It also allows fast application development and testing, particularly by small IT companies that cannot afford large investments on infrastructure. The most important advantage offered by clouds is in economies of scale; that is, when thousands of users share the same facility, the cost per user lessens. To offer such services, cloud computing encompasses many technologies and concepts such as virtualization, utility computing, a pay-as-you-go business model and scalable automatic provisioning on demand.

Clouds are deployed on physical infrastructure where cloud middleware is implemented for delivering services to customers. Such an infrastructure and middleware differ in their services, administrative domain and access to users. Therefore, the cloud deployments are classified mainly into three types: public cloud, private cloud and hybrid cloud (Figure 16.4).

16.2.3.1 Public Clouds

The public cloud is the most common deployment model where services are made avail able to anyone through the Internet. To support thousands of public domain users, data centres built by public cloud providers are quite large comprising thousands of servers with high-speed networks. Examples of popular public clouds are Amazon Web Services (AWS), Google AppEngine and Microsoft Azure. Public clouds are particularly attractive to customers with small enterprises or infrequent infrastructure usage, since these clouds provide a very good option to handle peak loads on the local infrastructure and for effective capacity planning. The fundamental characteristic of public clouds is its multitenancy, which is essentially achieved using sophisticated virtualization at various levels of the software stack. Being public clouds, QoS and security are the main issues that need to be ensured in their management.

16.2.3.2 Private Clouds

Private clouds are deployed within the premise of an organization to provide IT services to its internal users. Private cloud services offer greater control over the infrastructure, improving security and service resilience because its access is restricted to one or a few organizations. Such private deployment poses an inherent limitation to end user applications, that is, inability to scale elastically on demand as can be done using pubic cloud services.

16.2.3.3 Hybrid Clouds

Hybrid clouds are a deployment with the benefits of both public and private clouds. In this model, organizations deploy noncritical information and processing to the public cloud, whilst keeping critical services and data in their control. Therefore, organizations can utilize their (existing) in-house IT infrastructure for maintaining sensitive information within the premises, and can auto-scale their resources using public clouds for handling occasional peak loads. These resources or services are temporarily leased in peak load times and then released. The hybrid cloud, in general, applies to services related to IT infrastructure rather than software services.

16.2.1 Cloud Computing Characteristics

The key characteristics exhibited by clouds are shown in Figure 16.2 and discussed here:

• **Virtualized:** Resources (i.e. compute, storage and network capacity) in clouds are virtualized, and this is achieved at various levels including virtual machine (VM) and platform levels.

• **Service oriented:** Cloud is implemented using a service-oriented architecture model where all the capabilities or components are available over the network as a service. Whether it is software, a platform or infrastructure, everything is offered as a service.

• **Elastic:** Resources (i.e. compute, storage and network capacity) required for cloud applications can be dynamically provisioned and varied, that is, increased or decreased at runtime depending on user QoS requirements.

• **Dynamic and distributed:** Although cloud resources are virtualized, they are often distributed to enable the delivery of high-performance and/or reliable cloud services. These resources are flexible and can be adapted according to customer requirements such as software, network configuration and so on.

• **Shared (economy of scale):** Clouds are shared infrastructure where resources serve multiple customers with dynamic allocation according to their application’s demand. This sharing model is also termed the multi-tenant model. In general, customers neither have any direct control over physical resources nor they are aware of the resource location and with whom resources are being shared.

• **Market oriented (pay as you go):** In cloud computing, customers pay for services on a pay-per-use (or pay-as-you-go) basis. The pricing model can vary depending on the QoS expectation of application. This characteristic addresses the utility dimension of cloud computing. That means that cloud services are offered as ‘metered’services where providers have an accounting model for measuring the use of the services, which helps in the development of different pricing plans and models. The accounting model helps in the control and optimization of resource usage.

• **Autonomic:** To provide highly reliable services, clouds exhibit autonomic behaviour by managing themselves in case of failures or performance degradation.

Q1

B. What levels of GHG emissions are associated with the use of IT, past, present, and future?

Greenhouse Gas (GHG) Emissions Associated with IT: Past, Present, and Future

The information technology (IT) sector contributes significantly to global greenhouse gas (GHG) emissions due to the energy-intensive nature of data centers, electronic devices, and related infrastructure. Understanding the levels of GHG emissions over time helps identify trends, challenges, and opportunities for mitigating IT's environmental impact.

Past: GHG Emissions in IT

Levels and Trends:

In the early 2000s, IT-related GHG emissions were relatively low compared to other sectors. However, the rapid adoption of computers, data centers, and internet services began increasing energy demands. By 2007, IT was estimated to account for 2% of global GHG emissions, equivalent to the aviation industry's footprint at the time.

Key Contributors:

Data Centers: Inefficient cooling systems and outdated servers consumed significant energy.

Devices: Widespread adoption of desktops, laptops, and mobile phones led to higher electricity usage.

Manufacturing: Producing hardware involved energy-intensive processes that emitted CO_2.

Challenges:

Limited awareness of energy-efficient practices.

Predominance of non-renewable energy sources in powering IT infrastructure.

Present: GHG Emissions in IT

Levels and Trends:

As of the 2020s, IT contributes approximately 3-4% of global GHG emissions. This increase is attributed to the exponential growth in cloud computing, artificial intelligence (AI), and the Internet of Things (IoT). However, advancements in energy efficiency and renewable energy usage have slowed the growth rate of emissions.

Key Contributors:

Data Centers: While more efficient, the sheer scale of operations has grown. Large hyperscale data centers consume vast amounts of energy, though many are now powered by renewable sources.

Network Infrastructure: The expansion of 5G networks and global connectivity increases energy demands.

Consumer Electronics: The proliferation of smart devices and IoT adds to energy consumption.

Mitigation Efforts:

Companies like Google, Microsoft, and Amazon have committed to carbon neutrality and use renewable energy for operations.

Advances in hardware design, such as energy-efficient processors, reduce device-level energy consumption.

AI and machine learning optimize energy use in data centers.

Future: Projected GHG Emissions in IT

Levels and Trends:

By 2040, IT's GHG emissions could rise to 6-8% of global emissions if current growth trends continue without intervention. This would make IT one of the largest contributors to global emissions.

Drivers of Growth:

AI and Big Data: These technologies require high computational power, increasing energy demand.

Digital Transformation: As more industries digitize, IT infrastructure will expand significantly.

Edge Computing and IoT: Billions of connected devices will further escalate energy usage.

Mitigation Potential:

Increased reliance on renewable energy sources for IT operations.

Development of carbon capture technologies integrated into data centers.

Innovations in quantum computing and other energy-efficient computing paradigms.

C. Describe a high-capacity modern data center and compare this to a typical legacy data center.

Modern High-Capacity Data Centers vs. Legacy Data Centers

Data centers are the backbone of IT infrastructure, providing computational resources, storage, and connectivity for businesses and digital services. Over the years, modern data centers have evolved to meet growing demands for efficiency, scalability, and sustainability.

High-Capacity Modern Data Centers

Definition:

Modern data centers are advanced facilities designed to handle massive amounts of data, computation, and storage with high efficiency, scalability, and environmental sustainability.

Key Features:

Scalability and Capacity:

Support for hyperscale operations, serving global cloud providers like AWS, Microsoft Azure, and Google Cloud.

Modular designs allow for rapid expansion to meet growing demand.

Energy Efficiency:

Use renewable energy sources such as solar, wind, or hydroelectric power.

Advanced cooling systems like liquid cooling or free-air cooling reduce energy consumption.

Optimized for low Power Usage Effectiveness (PUE), often below 1.2.

Advanced Technologies:

Software-defined infrastructure for dynamic resource allocation.

AI and machine learning to optimize workloads, energy usage, and predictive maintenance.

High-speed network infrastructure with fiber-optic connectivity.

Sustainability:

Carbon-neutral or carbon-negative goals with renewable energy and carbon offset projects.

Recycling and reuse of waste heat in nearby facilities or urban heating networks.

Example: Google's hyperscale data centers employ AI to manage cooling systems, achieving up to a 30% reduction in energy consumption. These facilities are powered by 100% renewable energy.

Legacy Data Centers

Definition:

Legacy data centers refer to older, traditional facilities that lack the advanced technologies and efficiencies of modern systems. These centers were built to meet the IT demands of earlier decades.

Key Features:

Limited Capacity and Scalability:

Smaller physical and computational capacity compared to modern centers.

Expanding these centers often requires significant manual intervention and infrastructure upgrades.

Inefficient Energy Usage:

Use of outdated cooling systems, such as standard air conditioning.

Higher PUE ratios, often above 2.0, resulting in significant energy wastage.

Basic Technologies:

Hardware-centric designs with limited automation.

Relatively slower network speeds and less robust connectivity.

Lack of virtualization and software-defined capabilities.

Environmental Impact:

Often rely on grid power derived from non-renewable energy sources.

Inefficient cooling and energy systems lead to a larger carbon footprint.

Example: A typical enterprise data center built in the 2000s might house standalone servers running on older air-cooled systems with no energy optimization features.

D. What is a Smart Grid, and what are its features? How does it help in improving environmental sustainability?

Electrical power grids have now become vital for our existence, growth and development, and our total electrical power consumption continues to increase significantly – worldwide electric power consumption is expected to triple by 2050. Most of the existing power grids are over 50 years old and suffer from several problems. Their average energy efficiency is poor (about 33%), wasting energy and polluting the environment. They are also unreliable, incompatible to integrate with renewable energy sources and vulnerable to cyber-attack. Smart Grids intend to modernize the power grids by using IT and other technologies to reduce the transmission and distribution loss. It will balance the electricity supply and demand and improve the grid reliability by effectively monitoring and controlling the grid. It will also assist in optimizing resource use, and hence in enhancing environmental sustainability.

A Smart Grid is a digitally enabled power grid that gathers, distributes and acts on information about the supply and demand and the behaviour of all participants (suppliers and consumers) in order to improve the efficiency, reliability, economics and sustainability of electricity services (Wikipedia, 2012). A Smart Grid overlays an electric grid system with sensors, controls and wireless digital communication devices. Deployment of Smart Grids calls for customers to install special smart meters at their premises (some customers are, however, not willing to pay for new equipment). Much of the SmartGrid, including devices at the customers' premises, are computer controlled and interactive. A modern Smart Grid has the following features:

1. Runs more efficiently.

2. Provides higher quality power and minimizes power outages.

3. Accommodates all types of power sources and generation as well as storage options.

4. Motivates consumers to actively participate in operations of the grid.

5. Resists cyber-attacks and computer viruses.

6. Is capable of self-healing.

Smart Grids offer several operational, economic and environmental benefits such as improved efficiency, cost-effectiveness, less pollution, lower bills and fewer outages. In addition, smart grids also offer following key benefits:

• **Better integration with other energy sources:** As we seek to implement energy con servation measures and get energy from renewable resources such as solar and wind power, a Smart Grid would help make it easier to integrate photovoltaic, wind and other intermittent power sources.

• **Less power outages:** A Smart Grid uses real-time data to detect and isolate faults and to reconfigure the distribution network to minimize power outages to customers. It can anticipate equipment failures and reroute electric power autonomously to provide customers power without interruption.

• **Resilient:** A Smart Grid is more resilient and better able to handle peak power demand as wells as severe weather conditions.

• **Smarter consumption:** A Smart Grid helps consumers to change their behaviour around variable power price or billing rates. Customers could save money by shutting off appliances and using less power at times when electricity is most expensive. IT plays key roles in enabling Smart Grids through optimization, estimation and better monitoring and control. Cloud-based Smart Grid services have begun to emerge.

Q2

A. What are the global initiatives taken by different countries to solve the e-waste problem? Also, write and explain about different organizations working on it globally.

Global Initiatives and Organizations Addressing the E-Waste Problem

The increasing accumulation of electronic waste (e-waste) has become a significant global challenge due to its environmental and health impacts. Various countries and organizations have implemented policies, programs, and initiatives to address the issue effectively.

Global Initiatives by Countries

1. European Union (EU) – Waste Electrical and Electronic Equipment (WEEE) Directive

Description:

The WEEE Directive requires manufacturers to manage the disposal and recycling of electronic products. It sets targets for the collection, recovery, and recycling of e-waste.

Impact:

Member states achieved over 40% e-waste collection rates.

Promotes eco-design, making devices easier to recycle.

2. Japan – Home Appliance Recycling Law

Description:

Japan mandates the recycling of large household appliances, such as televisions, air conditioners, and refrigerators. Manufacturers bear the responsibility for recycling.

Impact:

High recycling rates for appliances.

Establishment of advanced recycling facilities.

3. United States – State-Level E-Waste Laws

Description:

While no federal law exists, individual states like California have e-waste recycling programs. The Electronic Waste Recycling Act in California imposes fees on electronic products to fund recycling programs.

Impact:

Encourages proper disposal and recycling of e-waste.

Supports the growth of local recycling industries.

4. India – E-Waste (Management) Rules

Description:

India's e-waste rules focus on Extended Producer Responsibility (EPR), requiring manufacturers to collect and recycle e-waste.

Impact:

Promotes formal recycling sectors.

Improves awareness and compliance among consumers and businesses.

5. China – Regulation for the Recycling of Waste Electric and Electronic Products

Description:

China implemented laws requiring producers to fund recycling efforts. Recycling facilities are integrated into urban planning.

Impact:

Reduction in informal recycling sectors.

Advances in e-waste processing technologies.

Global Organizations Working on E-Waste Management

1. Basel Convention

Overview:

A global treaty aimed at controlling the transboundary movement of hazardous waste, including e-waste.

Key Activities:

Prevents the illegal dumping of e-waste in developing countries.

Encourages safe disposal and recycling practices.

Impact:

Reduces the environmental burden on low-income countries.

Promotes global cooperation in hazardous waste management.

2. United Nations University (UNU)

Overview:

UNU conducts research and provides data on global e-waste trends and solutions.

Key Activities:

Publishes the Global E-Waste Monitor to track e-waste generation and recycling rates.

Develops training programs for governments and businesses.

Impact:

Raises awareness about e-waste issues.

Informs policy-making through reliable data.

3. International Telecommunication Union (ITU)

Overview:

ITU sets standards and promotes collaboration to reduce the environmental impact of ICT.

Key Activities:

Advocates for sustainable production and recycling of electronic devices.

Supports green ICT initiatives in developing countries.

Impact:

Improves global standards for e-waste management.

Encourages private-sector participation in recycling.

4. StEP (Solving the E-Waste Problem) Initiative

Overview:

A global coalition of academia, industry, and governments addressing e-waste.

Key Activities:

Promotes sustainable e-waste recycling technologies.

Facilitates international cooperation on policy and research.

Impact:

Advances eco-design and recycling best practices.

Enhances global dialogue on e-waste challenges.

5. World Economic Forum (WEF) – Circular Electronics Initiative

Overview:

WEF promotes circular economy principles for electronic products.

Key Activities:

Encourages repair, reuse, and recycling of electronics.

Engages stakeholders across the supply chain to reduce e-waste.

Impact:

Fosters innovation in circular design.

Reduces resource consumption and waste generation.

Challenges in Global E-Waste Management

Lack of Awareness: Many consumers are unaware of proper disposal methods.

Informal Recycling: In developing countries, e-waste is often processed informally, leading to environmental and health hazards.

Enforcement Issues: Ineffective enforcement of e-waste regulations limits their impact.

Q3

A. What is meant by Green IT? Explain Green Supply Chain and Logistics.

Green IT (Green Information Technology) refers to the practice of designing, manufacturing, using, and disposing of IT hardware, software, and infrastructure in a way that minimizes environmental impact. The focus is on reducing energy consumption, lowering greenhouse gas emissions, and promoting sustainable practices throughout the IT lifecycle.

A supply chain is a set of organizations directly linked by one or more of the upstream and downstream flows of products, services, finances and information from a source to a customer (Mentzer *et al.*, 2001).
According to the Council of Supply Chain Management Professionals (CSCMP), *supply chain management* encompasses the planning and management of all activities involved in sourcing, procurement, conversion and logistics management. It includes the crucial components of coordination and collaboration with suppliers, intermediaries, third-party service providers and customers. Supply chain management integrates supply and demand management within and across companies.
Due to increasing concerns about environmental issues, organizations are required to effectively coordinate and manage their supply chain management with minimal impact on the environment.
Green supply chain management integrates environmental considerations into supply chain management including product design, material sourcing and selection, manufacturing processes, delivery of the final product to the consumers as well as end-of-life management of the product after its useful life (Srivastava, 2007; Wang and Gupta, 2011).
Driven by increasing customer awareness, regulatory concerns and the need to reduce the cost and environmental impact of transportation, organizations are adopting green supply chain management practices. For instance, Dell became carbon neutral in 2008 by improving its supply chain and logistics.
Texas Instruments achieved 20% annual savings by reducing its transit packaging budget for its semiconductor business through source reduction, recycling and use of reusable packaging systems.
Integrating environmental considerations into supply chain management has become an increasingly important issue for industry, government and academic researchers.

Supply chain managers are being required to respond to new legislation, standards and regulations; changing customer demands; and demand for improved efficiency, cost effectiveness and return on investment, all whilst simultaneously being ‘green’.
To green their supply chain, organization can adopt one or more of the following measures (Cognizant White Paper, 2008):

- Use IT and other technology solutions to facilitate green supply chain management
with a focus on end-to-end supply chain analysis and network design.
- Pay attention to reducing inventory and identifying optimal distribution solutions adopt
ing better information management for inventory and ordering of goods.
- Perform life cycle analysis for choosing products or solutions to minimize environmen
tal impact.
- Adopt industry standard frameworks like SCOR version 9 to identify potential areas
for green initiatives in the supply chain.
- Align green initiatives with the strategic objectives of the company.
- Follow green supply chain management best practices in implementing green initiatives.

B. Write down different techniques to minimize power usage in any organization.

Techniques to Minimize Power Usage in Organizations

Minimizing power usage in organizations not only reduces operational costs but also supports environmental sustainability by lowering carbon footprints. Below are various techniques to achieve power efficiency:

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1. Energy-Efficient Hardware

- **Adopt Energy Star-Certified Devices:** Use computers, monitors, and printers that meet energy efficiency standards.
- **Low-Power Servers and Processors:** Employ hardware designed for lower power consumption, such as ARM-based servers.
- **Solid-State Drives (SSDs):** Replace traditional hard drives with SSDs for faster, more energy-efficient performance.

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2. Virtualization and Cloud Computing

- **Server Virtualization:** Run multiple virtual servers on a single physical server to maximize resource utilization.
- **Cloud Solutions:** Shift workloads to energy-efficient cloud platforms that utilize renewable energy.
- **Application Consolidation:** Reduce the number of applications running simultaneously on servers.

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3. Power Management Settings

- **Enable Sleep and Hibernate Modes:** Automatically switch devices to low-power modes during inactivity.
- **Dynamic Voltage and Frequency Scaling (DVFS):** Adjust processor power dynamically based on workload requirements.
- **Monitor Power Usage:** Use tools to track and optimize energy consumption in real-time.

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4. Lighting Optimization

- **Switch to LED Lights:** Replace traditional incandescent or fluorescent bulbs with energy-efficient LED lighting.
- **Motion Sensors:** Install motion detectors to turn lights on or off based on occupancy.
- **Maximize Natural Lighting:** Design workspaces to use daylight effectively, reducing dependency on artificial lighting.

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5. Smart Cooling and Heating Systems

- **Optimize Data Center Cooling:**
  - Use AI-powered cooling systems.
  - Implement free-air cooling or liquid cooling technologies.
- **Smart Thermostats:** Use programmable thermostats to regulate building temperatures efficiently.
- **Heat Recovery Systems:** Reuse waste heat from data centers or equipment to warm buildings.

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6. Renewable Energy Integration

- **Solar Panels:** Install photovoltaic systems to generate on-site renewable energy.
- **Wind Turbines:** Use wind energy where feasible to supplement power needs.
- **Battery Storage Systems:** Store excess renewable energy for later use.

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7. Employee Awareness and Training

- **Encourage Energy-Saving Practices:** Educate employees about turning off devices and lights when not in use.
- **Green IT Policies:** Implement guidelines for energy-efficient behavior.
- **Incentives:** Offer rewards for employees who suggest or adopt power-saving ideas.

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8. Advanced IT Solutions

- **Thin Clients:** Use low-power thin clients instead of traditional desktops.
- **Edge Computing:** Reduce energy demands by processing data closer to its source, decreasing the load on central servers.
- **Power-Efficient Algorithms:** Optimize software to run tasks with minimal energy requirements.

9. Regular Maintenance

- **Equipment Cleaning:** Clean dust from fans and components to ensure efficient cooling and operation.
- **Replace Outdated Devices:** Upgrade older equipment with more energy-efficient models.
- **Audit Energy Usage:** Conduct regular audits to identify inefficiencies and address them.

10. Use Smart Grids and IoT Technology

- **Smart Grids:** Connect to energy-efficient power grids that balance supply and demand effectively.
- **IoT Sensors:** Deploy smart sensors to monitor and control energy usage in real-time.
- **Automated Systems:** Implement AI-driven systems for energy optimization across facilities.

11. Policies and Certifications

- **Green Certifications:** Aim for certifications like LEED (Leadership in Energy and Environmental Design) for buildings.
- **Carbon Neutral Goals:** Set organizational targets to reduce energy-related emissions.
- **Power Purchase Agreements (PPAs):** Collaborate with renewable energy providers to secure green power.

Conclusion

Implementing these techniques can significantly reduce power consumption in organizations. The integration of energy-efficient hardware, renewable energy sources, and employee awareness initiatives ensures sustainable operations while reducing costs and environmental impact. By adopting a combination of these strategies, organizations can achieve both economic and environmental goals

Q4

A. Describe Reuse, Recycle, and Dispose with real-life examples.

Whereas refurbishing and recycling are effective techniques for the end-of-life man agement of electronic devices, often they are not practical. The only option left at this stage would be disposal. There are a few known techniques of waste disposal that are commonly used in e-waste disposal as well. These are incineration, chemical decomposition and landfill. There is no single silver bullet that works in all cases for all materials or all kinds of e-waste. Incineration may result in the release of toxic gases and can pose a hazard. On the positive side, the heat generated may be used in generating electricity. Due to the nonbiodegradability of materials used in electronic devices, landfill may not be a viable option either. Chemical decomposition may be an option in some cases and will have to be executed in a managed manner. In short, safe disposal is a major problem and the solution depends on the specific material used in manufacturing the device.

Examples:

- **Landfill Disposal:** Non-recyclable waste, such as certain types of plastics or mixed materials, is sent to landfills.
- **Incineration:** Some waste, like medical or hazardous waste, is burned at high temperatures in incinerators to reduce its volume and potential harm.
- **E-Waste Disposal:** Batteries and electronics containing toxic chemicals are disposed of through specialized facilities to ensure that harmful substances like lead and mercury do not leach into the soil or water.

Benefits:

- Ensures harmful materials are managed safely.
- Helps maintain clean and hygienic surroundings.
- Protects ecosystems from contamination caused by improper waste management

Q5 A. How do enterprise resource planning systems integrate with greening enterprise activities? What are the challenges in integrating them?

A *sustainable enterprise* or *green enterprise* is a business that has no or minimal negative impact on the environment. A business can be designated as a green business if it meets the following four criteria (Cooney, 2009):
1. It incorporates principles of sustainability into each of its business decisions.
2. It supplies environmentally friendly products or services that replace the demand for nongreen products and/or services.
3. It is greener than its traditional competitors.
4. It has made an enduring commitment to environmental principles in its business operations.

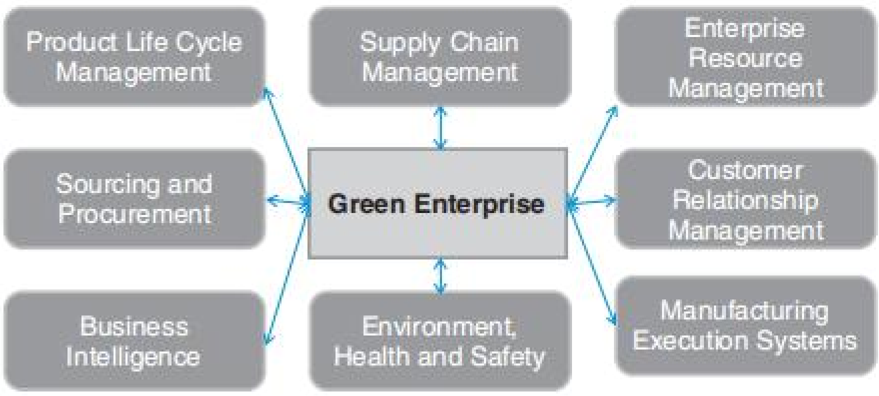

Figure 18.1 Making an enterprise greener.

A sustainable enterprise requires each of its functional units and activities to be green. The general guidelines for making an enterprise's functional units green are as follows:

• **Supply chain management:** Adopt techniques and plans to minimize inventory and wasted freight; adopt networked design using a carbon footprint.

• **Enterprise resource management:** Enable paperless transactions, and adopt techniques for workforce and parts optimization as well as intelligent device management.

• **Customer relationship management:** Adopt electronic means to maintain contact with customers.

• **Manufacturing execution systems:** Make proper plans for production scheduling, adopt co-product or by-product design and implement strategies for better yield management.

• **Environment, health and safety:** Monitor the environmental impact and carbon foot print, and comply with environmental, health and safety standards.

• **Business intelligence:** Include reporting systems for energy reporting and carbon foot printing.

• **Sourcing and procurement:** Select vendors by their sustainability rating.

• **Product life cycle management:** Design products that consume less energy; implement effective systems for product end-of-life management. Thus sustainability initiatives are taking many directions, and IT can and should play a major role in all these initiatives – greening *by* IT.